

<PROJECT PDF DUMP>
Generated at: 2026-04-23T16:43:28.466234 UTC
Root: C:\Users\morty\video-exchange-bot
Files included: 16

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FILE: .env
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```
BOT_TOKEN=8747618457:AAHz0LNqPr4wnSW0dDhT0xLwyREc-JKHINI
ADMINS=5272076117
WEBHOOK_BASE=https://placeholder.onrender.com
WEBHOOK_PATH=/webhook
DATABASE_URL=sqlite+aiosqlite:///bot.db
PORT=10000
```

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FILE: .env.example
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```
BOT_TOKEN=123456:ABC-DEF
ADMINS=123456789
WEBHOOK_BASE=https://your-app.onrender.com
WEBHOOK_PATH=/webhook
DATABASE_URL=postgresql+asyncpg://user:password@host:5432/dbname
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FILE: app\__init__.py
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FILE: app\admin_handlers.py
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```
from datetime import datetime, timedelta
from decimal import Decimal
from io import BytesIO

from aiogram import Router, F
from aiogram.filters import Command
from aiogram.fsm.state import State, StatesGroup
from aiogram.fsm.context import FSMContext
from aiogram.types import (
    Message, CallbackQuery,
    InlineKeyboardMarkup, InlineKeyboardButton
)
from sqlalchemy import select, func, desc

from app.config import ADMINS, OFFER_DEFAULT_RENT_COST_PER_DAY
from app.db import async_session
from app.models import (
    User, Video, Offer, BalanceLog, GameHistory,
    UserActionLog, OfferRental, OfferParticipation
)
from app.services import (
    get_user, get_user_by_id, get_user_by_username,
    get_user_dossier, update_user_balance, set_user_ban_status,
    count_pending_videos, count_approved_videos, count_rejected_videos,
    get_next_pending_video, approve_video, reject_video,
    get_admin_extended_stats, to_decimal, get_display_name,
    get_user_by_display_name, admin_create_offer,
    count_active_rentals, expire_old_rentals, log_user_action,
)
from app.keyboards import (
    admin_main_keyboard, moderation_keyboard,
    rejection_reason_keyboard, admin_after_action_keyboard,
    admin_offers_keyboard,
)

router = Router()

# =====
# STATES
# =====
class AdminUserState(StatesGroup):
    waiting_user_id = State()
    waiting_coins_amount = State()
    waiting_message_text = State()
    waiting_dossier_id = State()

class AdminManageState(StatesGroup):
    waiting_new_admin = State()
    waiting_remove_admin = State()

class AdminBroadcastState(StatesGroup):
    waiting_text = State()
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class AdminNicknameState(StatesGroup):
    waiting_user_id = State()
    waiting_new_nick = State()

class AdminOfferCreateState(StatesGroup):
    waiting_title = State()
    waiting_description = State()
    waiting_url = State()
    waiting_reward_preview = State()
    waiting_reward_final = State()
    waiting_rentable = State()
    waiting_rent_cost = State()
    waiting_max_rentals = State()

# =====
# HELPERS
# =====
def is_super_admin(tid: int) -> bool:
    return tid in ADMINS

async def check_admin(tid: int) -> bool:
    if tid in ADMINS:
        return True
    async with async_session() as session:
        user = await get_user(session, tid)
        if user and user.is_admin:
            return True
    return False

async def _safe_edit(callback: CallbackQuery, text: str, **kwargs):
    try:
        await callback.message.edit_text(text, **kwargs)
    except Exception:
        await callback.message.answer(text, **kwargs)

# =====
# ADMIN PANEL
# =====
@router.message(Command("admin"))
async def cmd_admin(message: Message):
    if not await check_admin(message.from_user.id):
        return
    sa = is_super_admin(message.from_user.id)
    await message.answer(
        "■ <b>■■■■■-■■■■■</b>",
        parse_mode="HTML",
        reply_markup=admin_main_keyboard(is_super=sa)
    )

@router.callback_query(F.data == "admin_center")
async def admin_center(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    sa = is_super_admin(callback.from_user.id)
    await _safe_edit(
        callback,
        "■ <b>■■■■■-■■■■■</b>",
        parse_mode="HTML",
        reply_markup=admin_main_keyboard(is_super=sa)
    )
    await callback.answer()

@router.callback_query(F.data == "admin_back")
async def admin_back(callback: CallbackQuery):
    await cmd_admin(callback.message)
    await callback.answer()

# =====
# QUEUE INFO
# =====
@router.callback_query(F.data == "admin_queue_info")
async def cb_queue(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    async with async_session() as session:
        p = await count_pending_videos(session)
        a = await count_approved_videos(session)

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await callback.answer()
```

```
@router.callback_query(F.data == "inv_suspicious_games")
async def inv_suspicious_games(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        week_ago = datetime.utcnow() - timedelta(days=7)
        rows = (await session.execute(
            select(
                User,
                func.count(GameHistory.id).label("games"),
                func.sum(GameHistory.result).label("profit"),
                func.sum(GameHistory.bet).label("total_bet"),
            )
            .join(GameHistory, GameHistory.user_id == User.id)
            .where(GameHistory.created_at >= week_ago)
            .group_by(User.id)
            .having(func.sum(GameHistory.result) > 50)
            .order_by(desc("profit"))
            .limit(20)
        )).all())

    if not rows:
        await callback.message.answer("■ ██████████ ████████ ■█ ████████.")
        await callback.answer()
        return

    text = "■ <b>████████████████████ ████████ (7 █████)</b>\n\n"
    for u, games, profit, total_bet in rows:
        text += (
            f"■ {get_display_name(u)} "  

            f"(<code>{u.telegram_id}</code>)\n"  

            f"   ████: {games} | ████████: {profit:.2f} | "  

            f"f"████████: {total_bet:.2f} | ████████: {u.balance:.2f}\n\n"
        )

    if len(text) > 4000:  

        text = text[:4000] + "\n..."

    kb = InlineKeyboardMarkup(inline_keyboard=[  

        [InlineKeyboardButton(text="■ ████████", callback_data="admin_investigation")]  

    ])
    await callback.message.answer(text, parse_mode="HTML", reply_markup=kb)  

    await callback.answer()
```

```
@router.callback_query(F.data == "inv_rich_detail")
async def inv_rich_detail(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        users = (await session.execute(
            select(User).order_by(desc(User.balance)).limit(15)
        )).scalars().all()

        text = "■ <b>■■■ ■■■■■■■■ (■■■■■■■■■)</b>\n\n"
        for i, u in enumerate(users, 1):
            admin_given = (await session.execute(
                select(func.sum(BalanceLog.amount)).where(
                    BalanceLog.user_id == u.id,
                    BalanceLog.source == "admin_balance",
                    BalanceLog.amount > 0
                )
            )).scalar_one() or Decimal("0")

            game_profit = (await session.execute(
                select(func.sum(GameHistory.result)).where(
                    GameHistory.user_id == u.id
                )
            )).scalar_one() or Decimal("0")

            text += (
                f"{i}. <b>{get_display_name(u)}</b> - {u.balance:.2f}\n"
                f"    ■■■■: {admin_given:.2f} | "
                f"■■■■: {game_profit:.2f} | "
                f"■■ <code>{u.telegram_id}</code>\n\n"
            )

        kb = InlineKeyboardMarkup(inline_keyboard=[
            [InlineKeyboardButton(text="■■■ ■■■■■■■■", callback_data="admin_investigation")]
        ])
        if len(text) > 4000:
```

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        text = text[:4000] + "\n..."
    await callback.message.answer(text, parse_mode="HTML", reply_markup=kb)
    await callback.answer()

@router.callback_query(F.data == "inv_balance_log")
async def inv_balance_log(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        rows = (await session.execute(
            select(BalanceLog, User)
            .join(User, User.id == BalanceLog.user_id)
            .order_by(desc(BalanceLog.created_at))
            .limit(30)
        )).all()

    if not rows:
        await callback.message.answer("■■■■ ■■■■.")
        await callback.answer()
        return

    text = "■ <b>■■■■■■■■■■ 30 ■■■■■■■■■■ ■■■■■■■■■■</b>\n\n"
    for log, u in rows:
        sign = "+" if log.amount >= 0 else ""
        text += (
            f"{log.created_at.strftime('%d.%m %H:%M')} | "
            f"<b>{get_display_name(u)}</b> | "
            f"{sign}{log.amount:.2f} [{log.source}]\n"
            f"    {log.balance_before:.2f} → {log.balance_after:.2f}"
        )
        if log.details:
            text += f"\n    ■ {log.details[:60]}"
        text += "\n\n"

    if len(text) > 4000:
        text = text[:4000] + "\n..."

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(text="■■ ■■■■", callback_data="admin_investigation")]
    ])
    await callback.message.answer(text, parse_mode="HTML", reply_markup=kb)
    await callback.answer()

@router.callback_query(F.data == "inv_export_ai")
async def inv_export_ai(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        rich_users = (await session.execute(
            select(User).order_by(desc(User.balance)).limit(10)
        )).scalars().all()

        sus_rows = (await session.execute(
            select(
                User,
                func.sum(GameHistory.result).label("profit"),
                func.count(GameHistory.id).label("games")
            )
            .join(GameHistory, GameHistory.user_id == User.id)
            .group_by(User.id)
            .having(func.sum(GameHistory.result) > 30)
            .order_by(desc("profit"))
            .limit(10)
        )).all()

        admin_rows = (await session.execute(
            select(BalanceLog, User)
            .join(User, User.id == BalanceLog.user_id)
            .where(BalanceLog.source == "admin_balance")
            .order_by(desc(BalanceLog.created_at))
            .limit(20)
        )).all()

        # ■■■■■■ - ■■■■■■■■■■, ■■■■ ■■■■■■■■ ■■■■■■■■■■
        try:
            rental_rows = (await session.execute(
                select(OfferRental, User)
                .join(User, User.id == OfferRental.renter_user_id)
                .order_by(desc(OfferRental.created_at))
                .limit(20)
            )).all()
        except Exception:

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        except Exception:
            pass

    await callback.message.answer(
        f"■ #{video_id} ■■■■■■■■■■. ■■■■■■■■■■: {reason_text}",
        reply_markup=admin_after_action_keyboard()
    )
    await callback.answer()

@router.callback_query(F.data == "admin_approve_all")
async def admin_approve_all(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    count = 0
    async with async_session() as session:
        while count < 100:
            video = await get_next_pending_video(session)
            if not video:
                break
            await approve_video(session, video.id)
            count += 1

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(text="■ ■ ■■■■■■■■■■", callback_data="admin_center")]
    ])
    await callback.message.answer(
        f"■ ■■■■■■■■■■: <b>{count}</b>",
        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

# =====
# USER MANAGEMENT
# =====
@router.callback_query(F.data == "admin_manage_users")
async def admin_manage_users(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(text="■ ■■■■■■■■", callback_data="admin_user_dossier"),
         [InlineKeyboardButton(text="■ ■■■■■■■■ ■■■■■■■■■■", callback_data="admin_give_coins"),
          [InlineKeyboardButton(text="■ ■■■■■■■■■■■■■■■■■■■", callback_data="admin_ban_user"),
           [InlineKeyboardButton(text="■ ■■■■■■■■■■■■■■■■■■■", callback_data="admin_unban_user"),
            [InlineKeyboardButton(text="■ ■■■■■■■■■■■■■■■■■■■", callback_data="admin_message_user"),
             [InlineKeyboardButton(text="■ ■■■■■■■■■■■■■■■■■■■", callback_data="admin_change_nickname"),
              [InlineKeyboardButton(text="■ ■■■■■■■■■■■■■■■■■■■", callback_data="admin_broadcast"),
               [InlineKeyboardButton(text="■ ■■■■■■■■", callback_data="admin_center")],
              ],
             ],
            ],
           ],
          ],
         ],
        ],
    ])
    await _safe_edit(
        callback,
        "■ <b>■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■</b>",
        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

# --- DOSSIER ---
@router.callback_query(F.data == "admin_user_dossier")
async def admin_user_dossier_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminUserState.waiting_dossier_id)
    await callback.message.answer(
        "■■■■■■■■ Telegram ID, @username ■■■■ ■■■■ ■■■■■■■■■■■■■■■■■■■:"
    )
    await callback.answer()

@router.message(AdminUserState.waiting_dossier_id)
async def process_dossier(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
        return
    query = message.text.strip()
    async with async_session() as session:
        user = None
        if query.startswith("@"):
            user = await get_user_by_username(session, query)
        elif query.isdigit():
            user = await get_user(session, int(query))
        else:
```

[illegible]

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    ]),
])

if len(text) > 4000:
    text = text[:4000] + "\n..."
await message.answer(text, parse_mode="HTML", reply_markup=kb)
await state.clear()

@router.callback_query(F.data.startswith("full_balance_log:"))
async def full_balance_log(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    user_id = int(callback.data.split(":")[1])

    async with async_session() as session:
        u = await get_user_by_id(session, user_id)
        if not u:
            await callback.answer("■ ■ ■ ■ ■ ■ ■ ■ ■ ■.", show_alert=True)
            return
        logs = (await session.execute(
            select(BalanceLog)
            .where(BalanceLog.user_id == user_id)
            .order_by(desc(BalanceLog.created_at))
            .limit(100)
        )).scalars().all()

    report = (
        f"=== ■ ■ ■ ■ ■ ■ ■ ■ ■ ■: {get_display_name(u)} "
        f"(tg={u.telegram_id}) ==\n"
        f"■ ■ ■ ■ ■ ■ ■ ■ ■ ■: {u.balance}\n\n"
    )
    for log in logs:
        sign = "+" if log.amount >= 0 else ""
        report += (
            f"{log.created_at.strftime('%Y-%m-%d %H:%M:%S')} | "
            f"{sign}{log.amount} | {log.source} | "
            f"{log.balance_before}→{log.balance_after}"
        )
        if log.admin_id:
            report += f" | admin={log.admin_id}"
        if log.details:
            report += f" | {log.details}"
        report += "\n"

    buf = BytesIO(report.encode("utf-8"))
    buf.name = f"balance_{u.telegram_id}.txt"
    await callback.message.answer_document(
        buf,
        caption=f"■ ■ ■ ■ ■ ■ ■ ■ ■ ■: {get_display_name(u)}"
    )
    await callback.answer()

# --- GIVE COINS ---
@router.callback_query(F.data == "admin_give_coins")
async def admin_give_coins_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminUserState.waiting_user_id)
    await state.update_data(action="give_coins")
    await callback.message.answer("■ ■ ■ ■ ■ ■ ■ ■ ■ ■ TG ID / @username / ■ ■ ■ ■:")
    await callback.answer()

@router.callback_query(F.data.startswith("give_coins_to:"))
async def give_coins_to_user(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    user_id = int(callback.data.split(":")[1])
    await state.set_state(AdminUserState.waiting_coins_amount)
    await state.update_data(target_user_id=user_id)
    await callback.message.answer(
        "■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ (+■ ■ ■ ■ ■ ■ ■ ■ ■ ■, -■ ■ ■ ■ ■ ■ ■ ■ ■ ■):"
    )
    await callback.answer()

@router.message(AdminUserState.waiting_user_id)
async def process_user_id_for_action(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
        return
    data = await state.get_data()
    action = data.get("action")
    query = message.text.strip()

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```
async with async_session() as session:
    user = None
    if query.startswith("@"):
        user = await get_user_by_username(session, query)
    elif query.isdigit():
        user = await get_user(session, int(query))
    else:
        user = await get_user_by_display_name(session, query)

    if not user:
        await message.answer("■ ██████████ ■█ ███████.")
        await state.clear()
        return

    await state.update_data(
        target_user_id=user.id,
        target_tg_id=user.telegram_id
    )

    if action == "give_coins":
        await state.set_state(AdminUserState.waiting_coins_amount)
        await message.answer("■■■■■■■■ ██████████ ████:")

    elif action == "ban":
        ok = await set_user_ban_status(
            session, user.id, True, message.from_user.id
        )
        await message.answer(
            f"■ {get_display_name(user)} ██████████." if ok else "■ ███████."
        )
        if ok:
            try:
                await message.bot.send_message(
                    user.telegram_id, "■ ■█ ██████████."
                )
            except Exception:
                pass
        await state.clear()

    elif action == "unban":
        ok = await set_user_ban_status(
            session, user.id, False, message.from_user.id
        )
        await message.answer(
            f"■ {get_display_name(user)} ██████████." if ok else "■ ███████."
        )
        if ok:
            try:
                await message.bot.send_message(
                    user.telegram_id, "■ ██████████ ███████."
                )
            except Exception:
                pass
        await state.clear()

    elif action == "message":
        await state.set_state(AdminUserState.waiting_message_text)
        await message.answer("■■■■■■■■ ███████ ██████████:")


@router.message(AdminUserState.waiting_coins_amount)
async def process_coins_amount(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
        return
    try:
        amount = Decimal(message.text.strip())
    except Exception:
        await message.answer("■ ██████████ ███████ (50 ■██ -10).")
        return

    data = await state.get_data()
    user_id = data.get("target_user_id")
    if not user_id:
        await message.answer("■ ███████: ██████████ ■█ ███████.")
        await state.clear()
        return

    async with async_session() as session:
        ok = await update_user_balance(
            session, user_id, amount, message.from_user.id
        )
        if ok:
            user = await get_user_by_id(session, user_id)
            name = get_display_name(user) if user else str(user_id)
            await message.answer(
                f"■ ██████████ ██████████!\n"
                f"■ {name}\n"
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        f"██████████: {'+' if amount > 0 else ''}{amount}\n"
        f"████████ ██████████: {user.balance if user else '???'}"
    )
    if user:
        try:
            await message.bot.send_message(
                user.telegram_id,
                f"████████ ██████████ ██████████ ██████████ ██████████: "
                f"{'+' if amount > 0 else ''}{amount} ██████████"
            )
        except Exception:
            pass
    else:
        await message.answer("████████████████████████████████████████.")
    await state.clear()

# --- BAN / UNBAN ---
@router.callback_query(F.data == "admin_ban_user")
async def admin_ban_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminUserState.waiting_user_id)
    await state.update_data(action="ban")
    await callback.message.answer("██████████ ID/██████/@username ████████████████████:")
    await callback.answer()

@router.callback_query(F.data == "admin_unban_user")
async def admin_unban_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminUserState.waiting_user_id)
    await state.update_data(action="unban")
    await callback.message.answer("██████████ ID/██████/@username ████████████████████:")
    await callback.answer()

@router.callback_query(F.data.startswith("ban_user:"))
async def ban_user_direct(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    user_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        ok = await set_user_ban_status(
            session, user_id, True, callback.from_user.id
        )
    user = await get_user_by_id(session, user_id)
    if ok:
        await callback.answer("████████████████████████████████████████!", show_alert=True)
        if user:
            try:
                await callback.bot.send_message(
                    user.telegram_id, "████████████████████████████████████████."
                )
            except Exception:
                pass
    else:
        await callback.answer("████████████████████.", show_alert=True)

@router.callback_query(F.data.startswith("unban_user:"))
async def unban_user_direct(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    user_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        ok = await set_user_ban_status(
            session, user_id, False, callback.from_user.id
        )
    user = await get_user_by_id(session, user_id)
    if ok:
        await callback.answer("████████████████████████████████████████!", show_alert=True)
        if user:
            try:
                await callback.bot.send_message(
                    user.telegram_id, "████████████████████████████████████████."
                )
            except Exception:
                pass
    else:
        await callback.answer("████████████████████.", show_alert=True)

```

```
# --- MESSAGE USER ---
@router.callback_query(F.data == "admin_message_user")
async def admin_message_user_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminUserState.waiting_user_id)
    await state.update_data(action="message")
    await callback.message.answer("██████ ID/███/@username ███████:")
    await callback.answer()

@router.message(AdminUserState.waiting_message_text)
async def process_message_text(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
        return
    data = await state.get_data()
    target_tg_id = data.get("target_tg_id")
    if not target_tg_id:
        await message.answer("■ █████: ██████████ █ █████.")
        await state.clear()
        return
    try:
        await message.bot.send_message(
            target_tg_id,
            f"■ <b>████████████████████████████████████████</b>\n\n{message.text}",
            parse_mode="HTML"
        )
        await message.answer("■ ███████████████████████████████████████!")
    except Exception as e:
        await message.answer(f"■ ███████████████████████████████████████: {e}")
        await state.clear()

# --- NICKNAME ---
@router.callback_query(F.data == "admin_change_nickname")
async def admin_change_nickname_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    await state.set_state(AdminNicknameState.waiting_user_id)
    await callback.message.answer("██████ ID/███/@username ███████:")
    await callback.answer()

@router.callback_query(F.data.startswith("admin_set_nick:"))
async def admin_set_nick_direct(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    user_id = int(callback.data.split(":")[1])
    await state.set_state(AdminNicknameState.waiting_new_nick)
    await state.update_data(target_user_id=user_id)
    await callback.message.answer(
        "████████████████████████████████████████ (████ '██████' ███████████████████):"
    )
    await callback.answer()

@router.message(AdminNicknameState.waiting_user_id)
async def admin_nick_process_user(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
        return
    query = message.text.strip()
    async with async_session() as session:
        user = None
        if query.startswith("@"):
            user = await get_user_by_username(session, query)
        elif query.isdigit():
            user = await get_user(session, int(query))
        else:
            user = await get_user_by_display_name(session, query)

    if not user:
        await message.answer("■ ███████████████████████████████████████.")
        await state.clear()
        return
    await state.update_data(target_user_id=user.id)

    await state.set_state(AdminNicknameState.waiting_new_nick)
    await message.answer(
        f"████████████████████████████████████████ {get_display_name(user)} (████ '██████'):"
    )

@router.message(AdminNicknameState.waiting_new_nick)
async def admin_nick_process_new(message: Message, state: FSMContext):
    if not await check_admin(message.from_user.id):
```



```

    ).scalars().all()

sent = failed = 0
for tg_id in tg_ids:
    try:
        await message.bot.send_message(
            tg_id,
            f"<b>██████████</b></b>\n\n{message.text}",
            parse_mode="HTML"
        )
        sent += 1
    except Exception:
        failed += 1

await message.answer(
    f"██████████ ██████████!\n██████████: {sent}\n██████████: {failed}"
)

# =====
# OFFERS MANAGEMENT (ADMIN)
# =====
@router.callback_query(F.data == "admin_offers_menu")
async def admin_offers_menu(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        pending_count = (await session.execute(
            select(func.count(Offer.id)).where(Offer.status == "pending")
        )).scalar_one()

        total_count = (await session.execute(
            select(func.count(Offer.id))
        )).scalar_one()

        active_count = (await session.execute(
            select(func.count(Offer.id)).where(
                Offer.is_active == True,
                Offer.status == "approved"
            )
        )).scalar_one()

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text=f"██████████ ██████████ (██████████)",
            callback_data="admin_create_offer"
        )],
        [InlineKeyboardButton(
            text=f"██████████ ██████████ ({pending_count})",
            callback_data="admin_offers_pending"
        )],
        [InlineKeyboardButton(
            text=f"██████████ ██████████ ({total_count})",
            callback_data="admin_offers_all"
        )],
        [InlineKeyboardButton(
            text=f"██████████ ██████████ ({active_count})",
            callback_data="admin_offers_active"
        )],
        [InlineKeyboardButton(
            text="██████████ ██████████",
            callback_data="admin_rentals_menu"
        )],
        [InlineKeyboardButton(text="██████████", callback_data="admin_center")]
    ])

    await _safe_edit(
        callback,
        f"██████████ <b>██████████ ██████████</b>\n\n"
        f"██████████: {total_count} | ██████████: {active_count} | "
        f"██████████ ██████████: {pending_count}",
        parse_mode="HTML",
        reply_markup=kb
    )

    await callback.answer()

# --- ██████████ ██████████ ██████████ (██████████) ---
@router.callback_query(F.data == "admin_create_offer")
async def admin_create_offer_start(callback: CallbackQuery, state: FSMContext):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    await state.set_state(AdminOfferCreateState.waiting_title)
    await callback.message.answer(
        "<b>██████████ ██████████ (██████████ 1/8)</b>\n\n"
```

[illegible]

[illegible]

```

rent_cost_per_day=data.get("rent_cost_per_day", Decimal("0")),
max_simultaneous_rentals=data.get("max_simultaneous_rentals", 0),
)

rentable_text = ""
if data.get("is_rentable"):
    rentable_text = (
        f"\n■ ■■■■■■: {data.get('rent_cost_per_day')} ■■■■■/■■■■\n"
        f"■■■■. ■■■■■■■■■■■■■■■■■■■: {data.get('max_simultaneous_rentals')}}"
    )

await message.answer(
    f"■ <b>■■■■■ ■■■■■!</b>\n\n"
    f"■ {data['title']}\n"
    f"■ ■■■■■. ■■■■■: {data['reward_preview']}\n"
    f"■ ■■■■■. ■■■■■: {data['reward_final']}"
    f"{rentable_text}\n\n"
    f"■■■■■ ■■■■■ ■■■■■■■ ■ ■■■■■ ■■■■■■■■■■■■■■■■■■■.",
    parse_mode="HTML"
)
await state.clear()

# --- ■■■■■ ■■■■■■ ■■ ■■■■■■■■ ---
@router.callback_query(F.data == "admin_offers_pending")
async def admin_offers_pending(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        offers = (await session.execute(
            select(Offer).where(Offer.status == "pending")
            .order_by(Offer.created_at)
        )).scalars().all()

    if not offers:
        await callback.message.answer(
            "■ ■■■ ■■■■■■ ■■ ■■■■■■■■.",
            reply_markup=InlineKeyboardMarkup(inline_keyboard=[
                [InlineKeyboardButton(text="■ ■■■■■", callback_data="admin_offers_menu")]
            ])
        )
        await callback.answer()
        return

    kb_buttons = []
    for offer in offers:
        kb_buttons.append([InlineKeyboardButton(
            text=f"■ {offer.title[:35]}",
            callback_data=f"admin_review_offer:{offer.id}"
        )])
    kb_buttons.append([InlineKeyboardButton(text="■ ■■■■■", callback_data="admin_offers_menu")])

    await callback.message.answer(
        f"■ <b>■■■■■ ■■ ■■■■■■■■ ({len(offers)})</b>",
        parse_mode="HTML",
        reply_markup=InlineKeyboardMarkup(inline_keyboard=kb_buttons)
    )
    await callback.answer()

# --- ■■■ ■■■■■■ ---
@router.callback_query(F.data == "admin_offers_all")
async def admin_offers_all(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        offers = (await session.execute(
            select(Offer).order_by(desc(Offer.created_at)).limit(25)
        )).scalars().all()

    if not offers:
        await callback.message.answer("■■■■■■■ ■■■.")
        await callback.answer()
        return

    text = "■ <b>■■■ ■■■■■■ (■■■■■■■■ 25)</b>\n\n"
    for o in offers:
        icon = "■" if o.is_active else ("■" if o.status == "pending" else "■")
        rent_icon = "■" if getattr(o, "is_rentable", False) else ""
        text += (
            f"{icon}{rent_icon} #{o.id} <b>{o.title[:30]}</b>\n"
            f"   ■■■■■■: {o.status} | "
            f"■■■■■■: {o.reward_preview}+{o.reward_final}\n"
        )

```

```

kb = InlineKeyboardMarkup(inline_keyboard=[
    [InlineKeyboardButton(text="■ ■ ■ ■ ■", callback_data="admin_offers_menu")]
])
if len(text) > 4000:
    text = text[:4000] + "\n..."
await callback.message.answer(text, parse_mode="HTML", reply_markup=kb)
await callback.answer()

# --- ■ ■ ■ ■ ■ ■ ■ ■ ---
@router.callback_query(F.data == "admin_offers_active")
async def admin_offers_active(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        offers = (await session.execute(
            select(Offer).where(
                Offer.is_active == True,
                Offer.status == "approved"
            ).order_by(desc(Offer.created_at))
        )).scalars().all()

    if not offers:
        await callback.message.answer(
            "■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■",
            reply_markup=InlineKeyboardMarkup(inline_keyboard=[
                [InlineKeyboardButton(text="■ ■ ■ ■ ■", callback_data="admin_offers_menu")]
            ])
        )
        await callback.answer()
        return

    kb_buttons = []
    for o in offers:
        rent_icon = "■" if getattr(o, "is_rentable", False) else ""
        kb_buttons.append([InlineKeyboardButton(
            text=f"■{rent_icon} #{o.id} {o.title[:30]}",
            callback_data=f"admin_review_offer:{o.id}"
        )])
    kb_buttons.append([InlineKeyboardButton(text="■ ■ ■ ■ ■", callback_data="admin_offers_menu")])

    await callback.message.answer(
        f"■ <b>■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ( {len(offers)} )</b>",
        parse_mode="HTML",
        reply_markup=InlineKeyboardMarkup(inline_keyboard=kb_buttons)
    )
    await callback.answer()

# --- ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ---
@router.callback_query(F.data.startswith("admin_review_offer:"))
async def admin_review_offer(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    offer_id = int(callback.data.split(":")[1])

    async with async_session() as session:
        offer = (await session.execute(
            select(Offer).where(Offer.id == offer_id)
        )).scalar_one_or_none()
        if not offer:
            await callback.answer("■ ■ ■ ■ ■ ■ ■", show_alert=True)
            return

    # ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
    participants = (await session.execute(
        select(func.count(OfferParticipation.id)).where(
            OfferParticipation.offer_id == offer_id
        )
    )).scalar_one()

    completed = (await session.execute(
        select(func.count(OfferParticipation.id)).where(
            OfferParticipation.offer_id == offer_id,
            OfferParticipation.status == "completed"
        )
    )).scalar_one()

    # ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
    active_rentals_count = 0
    try:
        active_rentals_count = (await session.execute(
            select(func.count(OfferRental.id)).where(
                OfferRental.offer_id == offer_id,

```

```

        OfferRental.status == "active"
    )
    ).scalar_one()
except Exception:
    pass

is_rentable = getattr(offer, "is_rentable", False)
rent_cost = getattr(offer, "rent_cost_per_day", 0)
max_rentals = getattr(offer, "max_simultaneous_rentals", 0)

text = (
    f"■ <b>■■■■■ </b>#{offer.id}</b>\n\n"
    f"■■■■■■■■■■: <b>{offer.title}</b>\n"
    f"■■■■■■■■■■: {offer.description}\n"
    f"URL: {offer.channel_url}\n"
    f"■■■■■■■■■■: <b>{offer.status}</b> | "
    f"■■■■■■■■■■: {'■' if offer.is_active else '■'}\n"
    f"■ ■■■■■■: {offer.reward_preview} + {offer.reward_final}\n"
    f"■■■■■■■■■■: {participants} | ■■■■■■■■■■: {completed}\n"
    f"■ ■■■■■■: {'■' if is_rentable else '■'}"
)
if is_rentable:
    text += (
        f"\n ■■■■: {rent_cost} ■■■■/■■■■\n"
        f"■■■■. ■■■■■■■■■■: {max_rentals}\n"
        f"■ ■■■■■■ ■■■■: {active_rentals_count}"
    )
text += f"\n■ ■■■■■■: {offer.created_at.strftime('%d.%m.%Y %H:%M')}"

# ■■■■■■ ■■■■■■■■■■
action_buttons = []
if offer.status == "pending":
    action_buttons.append([
        InlineKeyboardButton(
            text="■ ■■■■■■■■",
            callback_data=f"approve_offer:{offer_id}"
        ),
        InlineKeyboardButton(
            text="■ ■■■■■■■■",
            callback_data=f"reject_offer:{offer_id}"
        ),
    ])
else:
    toggle_text = "■ ■■■■■■■■■■■■■■■■■■" if offer.is_active else "■■ ■■■■■■■■■■■■■■■■■■"
    action_buttons.append([
        InlineKeyboardButton(
            text=toggle_text,
            callback_data=f"toggle_offer:{offer_id}"
        ),
    ])

action_buttons.append([
    InlineKeyboardButton(
        text="■ ■■■■■■■■",
        callback_data=f"delete_offer:{offer_id}"
    ),
])
action_buttons.append([
    InlineKeyboardButton(text="■■ ■■■■■■", callback_data="admin_offers_menu")
])

await callback.message.answer(
    text, parse_mode="HTML",
    reply_markup=InlineKeyboardMarkup(inline_keyboard=action_buttons)
)
await callback.answer()
```

```

@router.callback_query(F.data.startswith("approve_offer:"))
async def approve_offer_cb(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        offer = (await session.execute(
            select(Offer).where(Offer.id == offer_id)
        )).scalar_one_or_none()
        if not offer:
            await callback.answer("■■ ■■■■■■■■.", show_alert=True)
            return
        offer.status = "approved"
        offer.is_active = True
        await session.commit()

    if offer.creator_user_id:
        creator = await get_user_by_id(session, offer.creator_user_id)
        if creator:
```

[illegible]

```
@router.callback_query(F.data.startswith("confirm_delete_offer:"))
async def confirm_delete_offer_cb(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        offer = (await session.execute(
            select(Offer).where(Offer.id == offer_id)
        )).scalar_one_or_none()
        if not offer:
            await callback.answer("❌ Offer not found.", show_alert=True)
            return
        # Confirm deletion
        offer.is_active = False
        offer.status = "deleted"
        await session.commit()

    await callback.message.answer(f"❌ Offer #{offer_id} deleted.")
    await callback.answer()

# =====
# RENTALS MANAGEMENT
# =====
@router.callback_query(F.data == "admin_rentals_menu")
async def admin_rentals_menu(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
        return

    async with async_session() as session:
        try:
            active_count = await count_active_rentals(session)
            # =====
            recent_rentals = (await session.execute(
                select(OfferRental, User, Offer)
                .join(User, User.id == OfferRental.renter_user_id)
                .join(Offer, Offer.id == OfferRental.offer_id)
                .order_by(desc(OfferRental.created_at))
                .limit(10)
            )).all()
        except Exception:
            active_count = 0
            recent_rentals = []

    text = f"❌ <b>===== </b>\n\n=====: {active_count}\n\n"

    kb_buttons = []
    if recent_rentals:
        text += "<b>===== </b>\n"
        for rental, user, offer in recent_rentals:
            expires = rental.expires_at.strftime('%d.%m') if rental.expires_at else "???"
            text += (
                f"• {get_display_name(user)} → {offer.title[:20]}\n"
                f"  {rental.renter_channel_title[:25]} | "
                f"❌ {expires} | {rental.status}\n"
            )
            if rental.status == "pending":
                kb_buttons.append([InlineKeyboardButton(
                    text=f"❌ =====: {rental.renter_channel_title[:25]}",
                    callback_data=f"admin_review_rental:{rental.id}"
                )])

    kb_buttons.extend([
        [InlineKeyboardButton(
            text=f"❌ =====",
            callback_data="admin_expire_rentals"
        )],
        [InlineKeyboardButton(text=f"❌ =====", callback_data="admin_offers_menu")],
    ])

    if len(text) > 4000:
        text = text[:4000] + "\n..."

    await callback.message.answer(
        text, parse_mode="HTML",
        reply_markup=InlineKeyboardMarkup(inline_keyboard=kb_buttons)
    )
    await callback.answer()

@router.callback_query(F.data.startswith("admin_review_rental:"))
async def admin_review_rental(callback: CallbackQuery):
    if not await check_admin(callback.from_user.id):
        await callback.answer()
```


[illegible]


```

STARTING_BALANCE = _get_float("STARTING_BALANCE", 2.0)
WATCH_COST = _get_float("WATCH_COST", 1.0)
UPLOAD_REWARD = _get_float("UPLOAD_REWARD", 0.5)

REFERRAL_REWARD_INVITER = _get_float("REFERRAL_REWARD_INVITER", 2.0)
REFERRAL_REWARD_NEW_USER = _get_float("REFERRAL_REWARD_NEW_USER", 1.0)

STARS_TO_COINS_RATE = _get_float("STARS_TO_COINS_RATE", 2.0)

STARS_PACKAGES = {
    "pack_1": {"stars": 1, "coins": 2, "title": "2 ████████"},
    "pack_5": {"stars": 5, "coins": 10, "title": "10 ██████"},
    "pack_10": {"stars": 10, "coins": 20, "title": "20 ██████"},
    "pack_25": {"stars": 25, "coins": 50, "title": "50 ██████"},
    "pack_50": {"stars": 50, "coins": 100, "title": "100 ██████"},
}

MONEY_PACKAGES = {
    "money_99": {"amount": 9900, "coins": 250, "title": "250 ██████"},
    "money_199": {"amount": 19900, "coins": 600, "title": "600 ██████"},
    "money_499": {"amount": 49900, "coins": 1800, "title": "1800 ██████"},
}

OFFER_BROADCAST_INTERVAL_HOURS = _get_float("OFFER_BROADCAST_INTERVAL_HOURS", 2.5)

# === ████████ ===
LEVEL_XP_BASE = _get_int("LEVEL_XP_BASE", 100)
LEVEL_XP_MULTIPLIER = _get_float("LEVEL_XP_MULTIPLIER", 1.5)
XP_PER_WATCH = _get_int("XP_PER_WATCH", 5)
XP_PER_UPLOAD = _get_int("XP_PER_UPLOAD", 20)
XP_PER_RATING = _get_int("XP_PER_RATING", 2)
XP_PER_COMMENT = _get_int("XP_PER_COMMENT", 3)
XP_PER_REACTION = _get_int("XP_PER_REACTION", 1)
XP_PER_GAME = _get_int("XP_PER_GAME", 3)

# === VIP ===
VIP_PRICE_STARS = _get_int("VIP_PRICE_STARS", 50)
VIP_DURATION_DAYS = _get_int("VIP_DURATION_DAYS", 30)
VIP_BONUS_MULTIPLIER = _get_float("VIP_BONUS_MULTIPLIER", 3.0)
VIP_FREE_PHOTOS = True
VIP_WATCH_DISCOUNT = _get_float("VIP_WATCH_DISCOUNT", 0.5)

# === █████ ===
DICE_MIN_BET = _get_int("DICE_MIN_BET", 1)
DICE_MAX_BET = _get_int("DICE_MAX_BET", 50)

# === ████████ ===
DAILY_QUESTS = [
    {"type": "watch", "target": 3, "reward": 2, "desc": "██████████ 3 ██████"},
    {"type": "upload", "target": 1, "reward": 3, "desc": "██████████ 1 ██████"},
    {"type": "rate", "target": 3, "reward": 1, "desc": "██████████ 3 ██████"},
    {"type": "comment", "target": 2, "reward": 2, "desc": "██████████ 2 ████████████████"},
    {"type": "react", "target": 5, "reward": 1, "desc": "██████████ 5 ██████████"},
]

PREMIUM_DAILY_QUESTS = [
    {"type": "watch", "target": 10, "reward": 8, "desc": "VIP: ██████████ 10 ██████"},
    {"type": "comment", "target": 5, "reward": 5, "desc": "VIP: ██████████ 5 ████████████████"},
]

REACTION_TYPES = ["👍", "❤️👍", "👎", "👏", "👤"]

# === ██████████ ===
COMMENTS_PER_10_MIN = _get_int("COMMENTS_PER_10_MIN", 5)
COMMENT_MIN_INTERVAL_SEC = _get_int("COMMENT_MIN_INTERVAL_SEC", 15)

# === ████████████████████ ██████████ ===
WEEKLY_TOP1_REWARD = _get_float("WEEKLY_TOP1_REWARD", 25.0)
WEEKLY_TOP2_REWARD = _get_float("WEEKLY_TOP2_REWARD", 15.0)
WEEKLY_TOP3_REWARD = _get_float("WEEKLY_TOP3_REWARD", 10.0)

# === ████████████████████ ===
PIN_OFFER_COST = _get_float("PIN_OFFER_COST", 100.0)
BUMP_VIDEO_COST = _get_float("BUMP_VIDEO_COST", 25.0)

# === ██████ ===
NICKNAME_FIRST_FREE = True
NICKNAME_CHANGE_COST = _get_float("NICKNAME_CHANGE_COST", 50.0)
NICKNAME_MIN_LENGTH = 3
NICKNAME_MAX_LENGTH = 24

# === ██████████ / ██████████ ===
OFFER_DEFAULT_RENT_COST_PER_DAY = _get_float("OFFER_DEFAULT_RENT_COST_PER_DAY", 20.0)
OFFER_MIN_RENT_DAYS = _get_int("OFFER_MIN_RENT_DAYS", 1)
OFFER_MAX_RENT_DAYS = _get_int("OFFER_MAX_RENT_DAYS", 30)

=====
FILE: app\db.py

```

```
=====
from sqlalchemy.ext.asyncio import create_async_engine, async_sessionmaker, AsyncSession
```

```
from app.config import DATABASE_URL
from app.models import Base
```

```
def _fix_url(url: str) -> str:
    if url.startswith("postgres://"):
        return url.replace("postgres://", "postgresql+asyncpg://", 1)
    if url.startswith("postgresql://"):
        return url.replace("postgresql://", "postgresql+asyncpg://", 1)
    return url
```

```
engine = create_async_engine(
    _fix_url(DATABASE_URL),
    echo=False,
)
```

```
async_session = async_sessionmaker(
    engine,
    class_=AsyncSession,
    expire_on_commit=False,
)
```

```
async def init_db():
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.create_all)
```

```
async def reset_db():
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.drop_all)
        await conn.run_sync(Base.metadata.create_all)
```

```
=====
FILE: app\keyboards.py
=====
```

```
from aiogram.types import (
    InlineKeyboardMarkup, InlineKeyboardButton,
    ReplyKeyboardMarkup, KeyboardButton,
)
```

```
# =====
# ████████████████████
# =====
BTN_WATCH = "■ ██████████"
BTN_UPLOAD = "■ ██████████"
BTN_PROFILE = "■ ██████████"
BTN_BUY = "■ ████████"
BTN_OFFERS = "■ ████████"
BTN_REFERRALS = "■ ██████████"
BTN_BONUS = "■ ████████"
BTN_ADMIN = "■ ████████"
BTN_GAMES = "■ ████████"
BTN_TOPS = "■ ████████"
BTN_QUESTS = "■ ██████████"
BTN_VIP = "■ VIP"
BTN_LEVEL = "■ ██████████"
```

```
def main_menu(is_admin: bool = False) -> ReplyKeyboardMarkup:
    kb = [
        [KeyboardButton(text=BTN_WATCH), KeyboardButton(text=BTN_UPLOAD)],
        [KeyboardButton(text=BTN_PROFILE), KeyboardButton(text=BTN_GAMES)],
        [KeyboardButton(text=BTN_OFFERS), KeyboardButton(text=BTN_BUY)],
        [KeyboardButton(text=BTN_REFERRALS), KeyboardButton(text=BTN_BONUS)],
        [KeyboardButton(text=BTN_QUESTS), KeyboardButton(text=BTN_TOPS)],
        [KeyboardButton(text=BTN_LEVEL), KeyboardButton(text=BTN_VIP)],
    ]
    if is_admin:
        kb.append([KeyboardButton(text=BTN_ADMIN)])
    return ReplyKeyboardMarkup(keyboard=kb, resize_keyboard=True)
```

```
# =====
# ████████████████████
# =====
def admin_main_keyboard(is_super: bool = False) -> InlineKeyboardMarkup:
    buttons = [
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_queue_info"),
         InlineKeyboardButton(text="■ ██████████+", callback_data="admin_extended_stats")],
        [InlineKeyboardButton(text="■ ████████████████████", callback_data="admin_manage_users"),
         InlineKeyboardButton(text="■ ██████████", callback_data="admin_get_pending")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_approve_all"),
         InlineKeyboardButton(text="■ ██████████", callback_data="admin_offers_menu")],
    ]
```

```

        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_investigation")],
    ]
    if is_super:
        buttons.append([
            InlineKeyboardButton(
                text="■ ██████████ ██████████",
                callback_data="admin_manage_admins"
            )
        ])
    return InlineKeyboardMarkup(inline_keyboard=buttons)

def admin_offers_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(text="■ ██████████ ██████████", callback_data="admin_create_offer")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_offers_pending")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_offers_all")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_offers_active")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_rentals_menu")],
        [InlineKeyboardButton(text="■ ██████████", callback_data="admin_center")]
    ])

def moderation_keyboard(video_id: int) -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(
                text="■ ██████████",
                callback_data=f"mod_approve:{video_id}"
            ),
            InlineKeyboardButton(
                text="■ ██████████",
                callback_data=f"mod_reject:{video_id}"
            )
        ],
        [InlineKeyboardButton(
            text="■ ██████████",
            callback_data="admin_get_pending"
        )]
    ])

def rejection_reason_keyboard(video_id: int) -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ██████████",
            callback_data=f"reject_reason:{video_id}:duplicate"
        )],
        [InlineKeyboardButton(
            text="■ ██████████",
            callback_data=f"reject_reason:{video_id}:off_topic"
        )],
        [InlineKeyboardButton(
            text="■ ██████████ ██████████",
            callback_data=f"reject_reason:{video_id}:forbidden"
        )],
        [InlineKeyboardButton(
            text="■ ██████████",
            callback_data=f"reject_reason:{video_id}:other"
        )]
    ])

def admin_after_action_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ██████████",
            callback_data="admin_get_pending"
        )],
        [InlineKeyboardButton(
            text="■ ██████████-██████████",
            callback_data="admin_center"
        )]
    ])

# =====
# ████████████████████████████████████████
# =====
def rules_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ██████████ ██████████",
            callback_data="accept_rules"
        )]
    ])

```

```

def watch_choice_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ■■■■■■",
            callback_data="watch_video_content"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■",
            callback_data="watch_photo_content"
        )],
    ])

def video_rating_keyboard(video_id: int) -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(text="■ 1", callback_data=f"rate:{video_id}:1"),
            InlineKeyboardButton(text="■ 2", callback_data=f"rate:{video_id}:2"),
            InlineKeyboardButton(text="■ 3", callback_data=f"rate:{video_id}:3"),
            InlineKeyboardButton(text="■ 4", callback_data=f"rate:{video_id}:4"),
            InlineKeyboardButton(text="■ 5", callback_data=f"rate:{video_id}:5"),
        ],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■■■",
            callback_data=f"comments:{video_id}"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■",
            callback_data=f"reactions:{video_id}"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■",
            callback_data="watch_next"
        )],
    ])

def photo_actions_keyboard(photo_id: int) -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■",
            callback_data=f"reactions:{photo_id}"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■ ■■■■",
            callback_data="watch_next_photo"
        )],
    ])

def buy_coins_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ 2 ■■■■■■ (1 Star)",
            callback_data="buy:pack_1"
        )],
        [InlineKeyboardButton(
            text="■ 10 ■■■■■■ (5 Stars)",
            callback_data="buy:pack_5"
        )],
        [InlineKeyboardButton(
            text="■ 20 ■■■■■■ (10 Stars)",
            callback_data="buy:pack_10"
        )],
        [InlineKeyboardButton(
            text="■ 50 ■■■■■■ (25 Stars)",
            callback_data="buy:pack_25"
        )],
        [InlineKeyboardButton(
            text="■ 100 ■■■■■■ (50 Stars)",
            callback_data="buy:pack_50"
        )],
        [InlineKeyboardButton(
            text="=■ ■■■■ ■■■■■",
            callback_data="buy_custom"
        )],
    ])

def vip_buy_keyboard() -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ■■■■■■ VIP",
            callback_data="buy_vip"
        )],
    ])

```

[illegible]

```

[InlineKeyboardButton(text="■ ■■■ ■■■■■■■■", callback_data="top_viewers")],
[InlineKeyboardButton(text="■ ■■■ ■■ XP", callback_data="top_levels")],
[InlineKeyboardButton(text="■ ■■■ ■■■■■■■■", callback_data="top_richest")],
])

```

```

def quests_keyboard(quests: list) -> InlineKeyboardMarkup:
    buttons = []
    for q in quests:
        if q.completed and not q.reward_claimed:
            buttons.append([InlineKeyboardButton(
                text=f"■ ■■■■■■■■ ■■■■■■■■: {q.quest_type} ({q.reward} ■■■■■)",
                callback_data=f"quest_claim:{q.id}"
            )])
        elif q.completed:
            buttons.append([InlineKeyboardButton(
                text=f"■ {q.quest_type}: {q.progress}/{q.target} - ■■■■■■■■",
                callback_data=f"quest_done:{q.id}"
            )])
        else:
            buttons.append([InlineKeyboardButton(
                text=f"■ {q.quest_type}: {q.progress}/{q.target}",
                callback_data=f"quest_info:{q.id}"
            )])
    return InlineKeyboardMarkup(inline_keyboard=buttons)

def reaction_menu_keyboard(video_id: int) -> InlineKeyboardMarkup:
    return InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(text="■", callback_data=f"react:{video_id}:■"),
            InlineKeyboardButton(text="♥■", callback_data=f"react:{video_id}:♥■"),
            InlineKeyboardButton(text="■", callback_data=f"react:{video_id}:■"),
            InlineKeyboardButton(text="■", callback_data=f"react:{video_id}:■"),
            InlineKeyboardButton(text="■", callback_data=f"react:{video_id}:■"),
        ]
    ])

```

=====

FILE: app\logger.py

=====

```

import json
import logging
import traceback
from datetime import datetime

def get_logger(name: str) -> logging.Logger:
    return logging.getLogger(name)

def _make_payload(message: str, **kwargs) -> str:
    payload = {
        "time": datetime.utcnow().isoformat() + "Z",
        "message": message,
    }
    payload.update(kwargs)
    return json.dumps(payload, ensure_ascii=False)

def setup_logging(level: int = logging.INFO):
    root = logging.getLogger()
    root.setLevel(level)

    while root.handlers:
        root.handlers.pop()

    handler = logging.StreamHandler()
    handler.setLevel(level)
    handler.setFormatter(logging.Formatter("%(message)s"))
    root.addHandler(handler)

def log_info(logger: logging.Logger, message: str, **kwargs):
    logger.info(_make_payload(message, **kwargs))

def log_warning(logger: logging.Logger, message: str, **kwargs):
    logger.warning(_make_payload(message, **kwargs))

def log_error(logger: logging.Logger, message: str, **kwargs):
    logger.error(_make_payload(message, **kwargs))

def log_exception(logger: logging.Logger, message: str, **kwargs):
    kwargs["traceback"] = traceback.format_exc()
    logger.error(_make_payload(message, **kwargs))

```

```
def log_event(logger: logging.Logger, level: int, message: str, **kwargs):
    logger.log(level, _make_payload(message, **kwargs))
```

```
=====
FILE: app\main.py
=====
```

```
import asyncio
from aiohttp import web
from aiogram import Bot, Dispatcher
from aiogram.client.default import DefaultBotProperties
from aiogram.enums import ParseMode

from app.config import BOT_TOKEN, PORT
from app.db import engine, init_db
from app.user_handlers import router as user_router
from app.admin_handlers import router as admin_router
from app.logger import setup_logging, get_logger, log_info
```

```
setup_logging()
logger = get_logger(__name__)
```

```
async def on_startup(app):
    from app.migrate import main as run_migrations
    await init_db()
    try:
        await run_migrations()
    except Exception as e:
        log_info(logger, f"Migrations warning: {e}")
    log_info(logger, "Bot started, DB initialized")
```

```
async def main():
    if not BOT_TOKEN:
        raise RuntimeError("BOT_TOKEN is empty")

    bot = Bot(
        token=BOT_TOKEN,
        default=DefaultBotProperties(parse_mode=ParseMode.HTML),
    )
    dp = Dispatcher()
    dp.include_router(user_router)
    dp.include_router(admin_router)

    app = web.Application()
    app.on_startup.append(on_startup)
    runner = web.AppRunner(app)
    await runner.setup()
    site = web.TCPSite(runner, "0.0.0.0", int(PORT or 10000))
    await site.start()

    try:
        log_info(logger, "Polling started")
        await dp.start_polling(bot)
    finally:
        await runner.cleanup()
        await bot.session.close()
        await engine.dispose()
```

```
if __name__ == "__main__":
    asyncio.run(main())
```

```
=====
FILE: app\make_project_pdf.py
=====
```

```
import os
import logging
from pathlib import Path
from datetime import datetime

from reportlab.lib.pagesizes import A4
from reportlab.pdfbase.pdfmetrics import stringwidth
from reportlab.pdfgen import canvas

import sys
sys.path.insert(0, str(Path(__file__).parent.parent))
from app.logger import setup_logging, get_logger, log_info, log_warning, log_exception
from pathlib import Path
```

```
PROJECT_ROOT = Path(".")
OUTPUT_FILE = "project_dump.pdf"
```

```
MAX_PAGES = 90
MAX_SIZE_MB = 15
```

```

INCLUDE_EXTENSIONS = {
    ".py",
    ".txt",
    ".md",
    ".json",
    ".yaml",
    ".yml",
    ".toml",
    ".ini",
    ".env",
    ".sql",
}

SPECIAL_FILENAMES = {
    "Dockerfile",
    "Procfile",
    "requirements.txt",
    "runtime.txt",
    ".env",
}

EXCLUDED_DIRS = {
    ".git",
    ".venv",
    "venv",
    "__pycache__",
    "node_modules",
    ".idea",
    ".vscode",
    "dist",
    "build",
    ".pytest_cache",
    ".mypy_cache",
    ".ruff_cache",
    ".cache",
}

EXCLUDED_FILES = {
    OUTPUT_FILE,
}

MAX_FILE_SIZE_KB = 300

def is_text_candidate(path: Path) -> bool:
    if path.name in SPECIAL_FILENAMES:
        return True
    if path.suffix.lower() in INCLUDE_EXTENSIONS:
        return True
    if path.name.startswith(".env"):
        return True
    return False

def should_skip(path: Path) -> bool:
    if path.is_dir():
        return True

    if path.name in EXCLUDED_FILES:
        return True

    parts = set(path.parts)
    if parts & EXCLUDED_DIRS:
        return True

    if not is_text_candidate(path):
        return True

    try:
        if path.stat().st_size > MAX_FILE_SIZE_KB * 1024:
            return True
    except Exception:
        return True

    return False

def iter_project_files(root: Path):
    files = []
    for path in root.rglob("*"):
        if should_skip(path):
            continue
        files.append(path)
    files.sort(key=lambda p: str(p).lower())
    return files

```

```

def safe_read_text(path: Path) -> str:
    for enc in ("utf-8", "utf-8-sig", "cp1251", "latin-1"):
        try:
            return path.read_text(encoding=enc)
        except Exception:
            pass
    return "[[FAILED TO READ FILE]]"

def wrap_line(line: str, font_name: str, font_size: int, max_width: float):
    if not line:
        return [""]

    if stringWidth(line, font_name, font_size) <= max_width:
        return [line]

    parts = []
    current = ""

    for ch in line:
        test = current + ch
        if stringWidth(test, font_name, font_size) <= max_width:
            current = test
        else:
            if current:
                parts.append(current)
            current = ch

    if current:
        parts.append(current)

    return parts

def build_blocks(root: Path):
    files = iter_project_files(root)
    blocks = []

    header = [
        "<PROJECT PDF DUMP>",
        f"Generated at: {datetime.utcnow().isoformat()} UTC",
        f"Root: {root.resolve()}",
        f"Files included: {len(files)}",
        ""
    ]
    blocks.append("\n".join(header))

    for path in files:
        rel = path.relative_to(root)
        content = safe_read_text(path).rstrip()

        block = [
            "=" * 80,
            f"FILE: {rel}",
            "=" * 80,
            content,
            ""
        ]
        blocks.append("\n".join(block))

    return blocks, files

def render_pdf(blocks, output_path: str):
    page_width, page_height = A4

    configs = [
        {"font_size": 9, "line_gap": 11, "margin": 36},
        {"font_size": 8, "line_gap": 9, "margin": 26},
        {"font_size": 7, "line_gap": 8, "margin": 18},
        {"font_size": 6, "line_gap": 7, "margin": 14},
    ]

    best_result = None

    for cfg in configs:
        c = canvas.Canvas(output_path, pagesize=A4)
        font_name = "Courier"
        font_size = cfg["font_size"]
        line_gap = cfg["line_gap"]
        margin = cfg["margin"]

        usable_width = page_width - margin * 2
        y = page_height - margin
        pages = 1

        c.setFont(font_name, font_size)

```

```

def new_page():
    nonlocal y, pages
    c.showPage()
    c.setFont(font_name, font_size)
    y = page_height - margin
    pages += 1

for block in blocks:
    for raw_line in block.splitlines():
        wrapped = wrap_line(raw_line, font_name, font_size, usable_width)
        for line in wrapped:
            if y < margin:
                new_page()
            c.drawString(margin, y, line)
            y -= line_gap

        if y < margin:
            new_page()
        y -= line_gap

c.save()

size_bytes = os.path.getsize(output_path)
size_mb = size_bytes / (1024 * 1024)

best_result = {
    "pages": pages,
    "size_bytes": size_bytes,
    "size_mb": size_mb,
    "config": cfg,
}

if pages <= MAX_PAGES and size_mb <= MAX_SIZE_MB:
    return best_result

return best_result

def main():
    setup_logging()
    logger = get_logger(__name__)

    try:
        blocks, files = build_blocks(PROJECT_ROOT)
        result = render_pdf(blocks, OUTPUT_FILE)

        logger.info("=" * 60)
        logger.info(f"PDF created: {OUTPUT_FILE}")
        logger.info(f"Files included: {len(files)}")
        logger.info(f"Pages: {result['pages']}")
        logger.info(f"Size: {result['size_mb']:.2f} MB")
        logger.info(f"Config used: {result['config']}")
        logger.info("=" * 60)

        if result["pages"] > MAX_PAGES:
            logger.warning(f"PDF exceeds page limit ({result['pages']} > {MAX_PAGES})")

        if result["size_mb"] > MAX_SIZE_MB:
            logger.warning(f"PDF exceeds size limit ({result['size_mb']:.2f} MB > {MAX_SIZE_MB} MB)")
    except Exception:
        log_exception(logger, "Error during PDF generation")

if __name__ == "__main__":
    main()

=====
FILE: app\migrate.py
=====
import asyncio
import sys
sys.path.insert(0, '..')

from sqlalchemy import text
from app.db import engine
from app.logger import setup_logging, get_logger, log_info

setup_logging()
logger = get_logger(__name__)

MIGRATIONS = [
    # --- offers ---
    """
    ALTER TABLE offers
    ADD COLUMN IF NOT EXISTS created_by_user_id INTEGER NULL;
    """,
    """
    ALTER TABLE offers

```

```

ADD COLUMN IF NOT EXISTS offer_kind VARCHAR(30) DEFAULT 'system';
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS promotion_tier VARCHAR(30) DEFAULT 'basic';
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS payment_method VARCHAR(20) NULL;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS promo_price_coins NUMERIC(12,2) DEFAULT 0;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS promo_price_rub INTEGER DEFAULT 0;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS promo_price_stars INTEGER DEFAULT 0;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS priority_level INTEGER DEFAULT 10;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS is_featured BOOLEAN DEFAULT FALSE;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS moderation_comment TEXT NULL;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS approved_at TIMESTAMP NULL;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS is_rentable BOOLEAN DEFAULT FALSE;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS rent_cost_per_day NUMERIC(10,2) DEFAULT 0;
''''',
''''',
ALTER TABLE offers
ADD COLUMN IF NOT EXISTS max_simultaneous_rentals INTEGER DEFAULT 1;
''''',
''''',
# --- users ---
''''',
ALTER TABLE users
ADD COLUMN IF NOT EXISTS display_name VARCHAR(32) NULL;
''''',
''''',
ALTER TABLE users
ADD COLUMN IF NOT EXISTS nickname_set BOOLEAN DEFAULT FALSE;
''''',
''''',
# --- game_sessions ---
''''',
CREATE TABLE IF NOT EXISTS game_sessions (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES users(id),
    window_start TIMESTAMP NOT NULL,
    games_played INTEGER DEFAULT 0,
    paid_at TIMESTAMP NULL
);
''''',
''''',
CREATE INDEX IF NOT EXISTS ix_game_sessions_user_id
ON game_sessions (user_id);
''''',
''''',
# --- balance_logs ---
''''',
CREATE TABLE IF NOT EXISTS balance_logs (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES users(id),
    amount NUMERIC(10,2) NOT NULL,
    balance_before NUMERIC(10,2) NOT NULL,
    balance_after NUMERIC(10,2) NOT NULL,
    source VARCHAR(100) NOT NULL,
    source_id INTEGER NULL,
    admin_id INTEGER NULL,
    details TEXT NULL,
    created_at TIMESTAMP DEFAULT NOW()
);
''''',
''''',

```

```

"""
CREATE INDEX IF NOT EXISTS ix_balance_logs_user_id
ON balance_logs (user_id);
"""
"""
CREATE INDEX IF NOT EXISTS ix_balance_logs_created_at
ON balance_logs (created_at);
"""
"""
CREATE INDEX IF NOT EXISTS ix_balance_logs_source
ON balance_logs (source);
"""
# --- offer_rentals ---
"""
CREATE TABLE IF NOT EXISTS offer_rentals (
    id SERIAL PRIMARY KEY,
    offer_id INTEGER NOT NULL REFERENCES offers(id),
    renter_user_id INTEGER NOT NULL REFERENCES users(id),
    renter_channel_title VARCHAR(255) NOT NULL,
    renter_channel_url TEXT NOT NULL,
    rent_days INTEGER NOT NULL,
    cost_paid NUMERIC(10,2) NOT NULL,
    status VARCHAR(20) DEFAULT 'pending',
    created_at TIMESTAMP DEFAULT NOW(),
    expires_at TIMESTAMP NULL
);
"""
"""
CREATE INDEX IF NOT EXISTS ix_offer_rentals_offer_id
ON offer_rentals (offer_id);
"""
"""
CREATE INDEX IF NOT EXISTS ix_offer_rentals_renter_user_id
ON offer_rentals (renter_user_id);
"""
"""
CREATE INDEX IF NOT EXISTS ix_offer_rentals_status
ON offer_rentals (status);
"""
# --- ██████████ ---
"""
CREATE INDEX IF NOT EXISTS ix_offers_created_by_user_id
ON offers (created_by_user_id);
"""
"""
CREATE INDEX IF NOT EXISTS ix_offers_status
ON offers (status);
"""
"""
CREATE INDEX IF NOT EXISTS ix_game_history_user_id
ON game_history (user_id);
"""
"""
CREATE INDEX IF NOT EXISTS ix_game_history_created_at
ON game_history (created_at);
"""
]

async def main():
    async with engine.begin() as conn:
        for sql in MIGRATIONS:
            try:
                await conn.execute(text(sql.strip()))
            except Exception as e:
                # SQLite ██████████ PostgreSQL - ██████████
                log_info(logger, f"Migration skipped: {e}")
    await engine.dispose()
    log_info(logger, "Migrations applied successfully")

if __name__ == "__main__":
    asyncio.run(main())

=====
FILE: app\models.py
=====
from datetime import datetime
from decimal import Decimal
from sqlalchemy import (
    BigInteger, String, Numeric, Boolean,
    Integer, DateTime, ForeignKey, UniqueConstraint, Text, Date,
)
from sqlalchemy.orm import DeclarativeBase, Mapped, mapped_column, relationship

class Base(DeclarativeBase):
    pass

```

```

class User(Base):
    __tablename__ = "users"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    telegram_id: Mapped[int] = mapped_column(BigInteger, unique=True, nullable=False, index=True)
    username: Mapped[str | None] = mapped_column(String(255), nullable=True, index=True)
    display_name: Mapped[str | None] = mapped_column(String(32), nullable=True, index=True)
    first_name: Mapped[str | None] = mapped_column(String(255), nullable=True)
    last_name: Mapped[str | None] = mapped_column(String(255), nullable=True)
    phone_number: Mapped[str | None] = mapped_column(String(50), nullable=True)
    balance: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    status: Mapped[str] = mapped_column(String(50), default="active")
    is_admin: Mapped[bool] = mapped_column(Boolean, default=False)
    agreed_to_rules: Mapped[bool] = mapped_column(Boolean, default=False)
    nickname_set: Mapped[bool] = mapped_column(Boolean, default=False)
    referral_code: Mapped[str | None] = mapped_column(String(50), unique=True, nullable=True)
    referred_by_user_id: Mapped[int | None] = mapped_column(ForeignKey("users.id"), nullable=True)
    referral_earnings: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    last_bonus_at: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)
    xp: Mapped[int] = mapped_column(Integer, default=0)
    level: Mapped[int] = mapped_column(Integer, default=1)
    vip_until: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    videos: Mapped[list["Video"]] = relationship(back_populates="uploader")
    views: Mapped[list["VideoView"]] = relationship(back_populates="user")
    ratings: Mapped[list["VideoRating"]] = relationship(back_populates="user")
    payments: Mapped[list["Payment"]] = relationship(back_populates="user")
    comments: Mapped[list["Comment"]] = relationship(back_populates="user")
    reactions: Mapped[list["ContentReaction"]] = relationship(back_populates="user")
    game_history: Mapped[list["GameHistory"]] = relationship(back_populates="user")
    quest_progress: Mapped[list["DailyQuestProgress"]] = relationship(back_populates="user")
    game_sessions: Mapped[list["GameSession"]] = relationship(back_populates="user")
    action_logs: Mapped[list["UserActionLog"]] = relationship(back_populates="user")
    balance_logs: Mapped[list["BalanceLog"]] = relationship(back_populates="user")
    user_offers: Mapped[list["Offer"]] = relationship(back_populates="creator")
    rentals: Mapped[list["OfferRental"]] = relationship(
        back_populates="renter",
        foreign_keys="OfferRental.renter_user_id"
    )

class UserActionLog(Base):
    __tablename__ = "user_action_logs"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False, index=True)
    action: Mapped[str] = mapped_column(String(255), nullable=False)
    details: Mapped[str | None] = mapped_column(Text, nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="action_logs")

class BalanceLog(Base):
    """
    Баланс пользователя - это сумма денег, которую он имеет на своем счете.
    """
    __tablename__ = "balance_logs"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False, index=True)
    amount: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
    balance_before: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
    balance_after: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
    source: Mapped[str] = mapped_column(String(100), nullable=False)
    source_id: Mapped[int | None] = mapped_column(Integer, nullable=True)
    admin_id: Mapped[int | None] = mapped_column(Integer, nullable=True)
    details: Mapped[str | None] = mapped_column(Text, nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow, index=True)

    user: Mapped["User"] = relationship(back_populates="balance_logs")

class Video(Base):
    __tablename__ = "videos"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    uploader_user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    content_type: Mapped[str] = mapped_column(String(20), default="video")
    telegram_file_id: Mapped[str] = mapped_column(Text, nullable=False)
    telegram_file_unique_id: Mapped[str] = mapped_column(String(255), nullable=False, index=True)
    status: Mapped[str] = mapped_column(String(20), default="pending")
    duration_seconds: Mapped[int | None] = mapped_column(Integer, nullable=True)
    file_size: Mapped[int | None] = mapped_column(BigInteger, nullable=True)
    rejection_reason: Mapped[str | None] = mapped_column(String(255), nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

```

```

uploader: Mapped["User"] = relationship(back_populates="videos")
views: Mapped[list["VideoView"]] = relationship(back_populates="video")
ratings: Mapped[list["VideoRating"]] = relationship(back_populates="video")
comments: Mapped[list["Comment"]] = relationship(back_populates="video")
reactions: Mapped[list["ContentReaction"]] = relationship(back_populates="video")

```

```

class VideoView(Base):
    __tablename__ = "video_views"
    __table_args__ = (
        UniqueConstraint("user_id", "video_id", name="uq_user_video_view"),
    )

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    video_id: Mapped[int] = mapped_column(ForeignKey("videos.id"), nullable=False)
    watched_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="views")
    video: Mapped["Video"] = relationship(back_populates="views")

```

```

class VideoRating(Base):
    __tablename__ = "video_ratings"
    __table_args__ = (
        UniqueConstraint("user_id", "video_id", name="uq_user_video_rating"),
    )

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    video_id: Mapped[int] = mapped_column(ForeignKey("videos.id"), nullable=False)
    rating: Mapped[int] = mapped_column(Integer, nullable=False)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="ratings")
    video: Mapped["Video"] = relationship(back_populates="ratings")

```

```

class Comment(Base):
    __tablename__ = "comments"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    video_id: Mapped[int] = mapped_column(ForeignKey("videos.id"), nullable=False)
    text: Mapped[str] = mapped_column(Text, nullable=False)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="comments")
    video: Mapped["Video"] = relationship(back_populates="comments")

```

```

class ContentReaction(Base):
    __tablename__ = "content_reactions"
    __table_args__ = (
        UniqueConstraint("user_id", "video_id", name="uq_user_video_reaction"),
    )

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    video_id: Mapped[int] = mapped_column(ForeignKey("videos.id"), nullable=False)
    reaction_type: Mapped[str] = mapped_column(String(10), nullable=False)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="reactions")
    video: Mapped["Video"] = relationship(back_populates="reactions")

```

```

class Payment(Base):
    __tablename__ = "payments"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    payload: Mapped[str] = mapped_column(String(255), unique=True, nullable=False)
    stars_amount: Mapped[int] = mapped_column(Integer, nullable=False)
    coins_amount: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
    status: Mapped[str] = mapped_column(String(50), default="pending")
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="payments")

```

```

class Offer(Base):
    __tablename__ = "offers"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    creator_user_id: Mapped[int | None] = mapped_column(ForeignKey("users.id"), nullable=True)
    title: Mapped[str] = mapped_column(String(255), nullable=False)
    description: Mapped[str] = mapped_column(Text, nullable=False)

```

```

channel_url: Mapped[str] = mapped_column(Text, nullable=False)
reward_preview: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("5"))
reward_final: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("35"))
penalty_unsubscribe: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("40"))
is_active: Mapped[bool] = mapped_column(Boolean, default=True)
status: Mapped[str] = mapped_column(String(20), default="approved")
# #####
is_rentable: Mapped[bool] = mapped_column(Boolean, default=False)
rent_cost_per_day: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
max_simultaneous_rentals: Mapped[int] = mapped_column(Integer, default=1)
created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

creator: Mapped["User"] = relationship(back_populates="user_offers")
participations: Mapped[list["OfferParticipation"]] = relationship(back_populates="offer")
rentals: Mapped[list["OfferRental"]] = relationship(back_populates="offer")

```

```

class OfferParticipation(Base):
    __tablename__ = "offer_participations"
    __table_args__ = (
        UniqueConstraint("user_id", "offer_id", name="uq_user_offer"),
    )

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    offer_id: Mapped[int] = mapped_column(ForeignKey("offers.id"), nullable=False)
    status: Mapped[str] = mapped_column(String(50), default="started")
    reward_given: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)
    checked_at: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)

    offer: Mapped["Offer"] = relationship(back_populates="participations")

```

```

class OfferRental(Base):
    """##### """
    __tablename__ = "offer_rentals"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    offer_id: Mapped[int] = mapped_column(ForeignKey("offers.id"), nullable=False, index=True)
    renter_user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False, index=True)
    renter_channel_title: Mapped[str] = mapped_column(String(255), nullable=False)
    renter_channel_url: Mapped[str] = mapped_column(Text, nullable=False)
    rent_days: Mapped[int] = mapped_column(Integer, nullable=False)
    cost_paid: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
    status: Mapped[str] = mapped_column(String(20), default="pending")
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)
    expires_at: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)

    renter: Mapped["User"] = relationship(
        back_populates="rentals",
        foreign_keys=[renter_user_id]
    )
    offer: Mapped["Offer"] = relationship(back_populates="rentals")

```

```

class GameHistory(Base):
    __tablename__ = "game_history"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False, index=True)
    game_type: Mapped[str] = mapped_column(String(50), nullable=False)
    bet: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    result: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    details: Mapped[str | None] = mapped_column(Text, nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow, index=True)

    user: Mapped["User"] = relationship(back_populates="game_history")

```

```

class DailyQuestProgress(Base):
    __tablename__ = "daily_quest_progress"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False)
    quest_type: Mapped[str] = mapped_column(String(50), nullable=False)
    quest_date: Mapped[datetime] = mapped_column(Date, nullable=False)
    progress: Mapped[int] = mapped_column(Integer, default=0)
    target: Mapped[int] = mapped_column(Integer, nullable=False)
    reward: Mapped[Decimal] = mapped_column(Numeric(10, 2), default=Decimal("0"))
    completed: Mapped[bool] = mapped_column(Boolean, default=False)
    reward_claimed: Mapped[bool] = mapped_column(Boolean, default=False)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=datetime.utcnow)

    user: Mapped["User"] = relationship(back_populates="quest_progress")

```

```

class GameSession(Base):

```

```

__tablename__ = "game_sessions"

id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
user_id: Mapped[int] = mapped_column(ForeignKey("users.id"), nullable=False, index=True)
window_start: Mapped[datetime] = mapped_column(DateTime, nullable=False)
games_played: Mapped[int] = mapped_column(Integer, default=0)
paid_at: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)

user: Mapped["User"] = relationship(back_populates="game_sessions")

=====
FILE: app\services.py
=====
import uuid
from datetime import datetime, timedelta
from decimal import Decimal, ROUND_DOWN

from sqlalchemy import select, func, desc
from sqlalchemy.ext.asyncio import AsyncSession

from app.models import (
    User, Video, VideoView, VideoRating,
    Payment, Offer, OfferParticipation, OfferRental,
    Comment, ContentReaction, GameHistory,
    DailyQuestProgress, GameSession,
    UserActionLog, BalanceLog,
)
from app.config import (
    STARTING_BALANCE, WATCH_COST, UPLOAD_REWARD,
    REFERRAL_REWARD_INVITER, REFERRAL_REWARD_NEW_USER,
    STARS_PACKAGES, STARS_TO_COINS_RATE,
    WEEKLY_TOP1_REWARD, WEEKLY_TOP2_REWARD, WEEKLY_TOP3_REWARD,
    DAILY_QUESTS, PREMIUM_DAILY_QUESTS,
    BUMP_VIDEO_COST, PIN_OFFER_COST,
    NICKNAME_CHANGE_COST, NICKNAME_FIRST_FREE,
    NICKNAME_MIN_LENGTH, NICKNAME_MAX_LENGTH,
    OFFER_MIN_RENT_DAYS, OFFER_MAX_RENT_DAYS,
)

PHOTO_UPLOAD_REWARD = Decimal("0.1")
OFFER_STEP_1_REWARD = Decimal("5")
OFFER_PENALTY = Decimal("40")
GAME_SESSION_LIMIT = 10
GAME_SESSION_COST = Decimal("10")
GAME_SESSION_HOURS = 4

# =====
# ████████
# =====
def to_decimal(val) -> Decimal:
    return Decimal(str(val))

def round_coin(val: Decimal) -> Decimal:
    return val.quantize(Decimal("0.01"), rounding=ROUND_DOWN)

def get_display_name(user: "User") -> str:
    """██████████ ██████████ ██████████. """
    if user.display_name:
        return user.display_name
    if user.first_name:
        return user.first_name
    if user.username:
        return f"@{user.username}"
    return f"User#{user.telegram_id}"

# =====
# ████████████████
# =====
async def log_balance_change(
    session: AsyncSession,
    user: "User",
    amount: Decimal,
    source: str,
    source_id: int = None,
    admin_id: int = None,
    details: str = None,
):
    """██████████ ██████████ ██████████ ██████████. """
    log = BalanceLog(
        user_id=user.id,
        amount=amount,
        balance_before=user.balance,
        balance_after=user.balance + amount,
        source=source,

```

```

        source_id=source_id,
        admin_id=admin_id,
        details=details,
    )
    session.add(log)

async def log_user_action(
    session: AsyncSession,
    user_id: int,
    action: str,
    details: str = None
):
    log = UserActionLog(user_id=user_id, action=action, details=details)
    session.add(log)
    try:
        await session.commit()
    except Exception:
        await session.rollback()

# =====
# ██████████
# =====
async def get_user(session: AsyncSession, telegram_id: int) -> "User | None":
    return (await session.execute(
        select(User).where(User.telegram_id == telegram_id)
    )).scalar_one_or_none()

async def get_user_by_id(session: AsyncSession, user_id: int) -> "User | None":
    return (await session.execute(
        select(User).where(User.id == user_id)
    )).scalar_one_or_none()

async def get_user_by_username(session: AsyncSession, username: str) -> "User | None":
    if username.startswith("@"):
        username = username[1:]
    return (await session.execute(
        select(User).where(User.username == username)
    )).scalar_one_or_none()

async def get_user_by_display_name(session: AsyncSession, display_name: str) -> "User | None":
    return (await session.execute(
        select(User).where(User.display_name == display_name)
    )).scalar_one_or_none()

async def get_or_create_user(
    session: AsyncSession,
    telegram_id: int,
    username: str = None,
    first_name: str = None,
    last_name: str = None,
    referral_code: str = None,
) -> tuple["User", bool]:
    user = await get_user(session, telegram_id)
    if user:
        user.username = username
        user.first_name = first_name
        user.last_name = last_name
        await session.commit()
        return user, False

    referred_by = None
    starting_bonus = to_decimal(STARTING_BALANCE)

    if referral_code:
        inviter = (await session.execute(
            select(User).where(User.referral_code == referral_code)
        )).scalar_one_or_none()
        if inviter and inviter.telegram_id != telegram_id:
            referred_by = inviter.id
            await log_balance_change(
                session, inviter,
                to_decimal(REFERRAL_REWARD_INVITER),
                "referral_inviter",
                details=f"New user tg_id={telegram_id}"
            )
            inviter.balance += to_decimal(REFERRAL_REWARD_INVITER)
            inviter.referral_earnings += to_decimal(REFERRAL_REWARD_INVITER)
            starting_bonus += to_decimal(REFERRAL_REWARD_NEW_USER)

    user = User(
        telegram_id=telegram_id,
        username=username,

```

[illegible]

```

# ##### / #####
# =====
async def approve_video(session: AsyncSession, video_id: int) -> "Video | None":
    v = (await session.execute(
        select(Video).where(Video.id == video_id)
    )).scalar_one_or_none()
    if not v or v.status != "pending":
        return v
    v.status = "approved"
    uploader = await get_user_by_id(session, v.uploader_user_id)
    if uploader:
        reward = (
            PHOTO_UPLOAD_REWARD
            if v.content_type == "photo"
            else to_decimal(UPLOAD_REWARD)
        )
        await log_balance_change(
            session, uploader, reward,
            "upload_approved", source_id=v.id
        )
        uploader.balance += reward
        await log_user_action(
            session, uploader.id,
            "video_approved",
            f"video_id={v.id}, reward={reward}"
        )
    await session.commit()
    return v

async def reject_video(
    session: AsyncSession,
    video_id: int,
    reason: str
) -> "Video | None":
    v = (await session.execute(
        select(Video).where(Video.id == video_id)
    )).scalar_one_or_none()
    if not v:
        return None
    v.status = "rejected"
    v.rejection_reason = reason
    await session.commit()
    await log_user_action(
        session, v.uploader_user_id,
        "video_rejected",
        f"video_id={v.id}, reason={reason}"
    )
    return v

async def get_next_pending_video(session: AsyncSession) -> "Video | None":
    return (await session.execute(
        select(Video)
        .where(Video.status == "pending")
        .order_by(Video.created_at)
        .limit(1)
    )).scalar_one_or_none()

async def count_pending_videos(session: AsyncSession) -> int:
    return (await session.execute(
        select(func.count(Video.id)).where(Video.status == "pending")
    )).scalar_one()

async def count_approved_videos(session: AsyncSession) -> int:
    return (await session.execute(
        select(func.count(Video.id)).where(Video.status == "approved")
    )).scalar_one()

async def count_rejected_videos(session: AsyncSession) -> int:
    return (await session.execute(
        select(func.count(Video.id)).where(Video.status == "rejected")
    )).scalar_one()

async def save_video(
    session: AsyncSession,
    user_id: int,
    file_id: str,
    file_unique_id: str,
    duration: int = None,
    file_size: int = None,
) -> "Video":
    video = Video(
        uploader_user_id=user_id,

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        content_type="video",
        telegram_file_id=file_id,
        telegram_file_unique_id=file_unique_id,
        duration_seconds=duration,
        file_size=file_size,
        status="pending",
    )
    session.add(video)
    await session.commit()
    await log_user_action(session, user_id, "upload_video", f"file={file_unique_id}")
    return video

async def save_photo(
    session: AsyncSession,
    user_id: int,
    file_id: str,
    file_unique_id: str,
    file_size: int = None,
) -> "Video":
    photo = Video(
        uploader_user_id=user_id,
        content_type="photo",
        telegram_file_id=file_id,
        telegram_file_unique_id=file_unique_id,
        file_size=file_size,
        status="pending",
    )
    session.add(photo)
    await session.commit()
    await log_user_action(session, user_id, "upload_photo", f"file={file_unique_id}")
    return photo

async def get_random_video_for_user(
    session: AsyncSession,
    user_id: int
) -> "Video | None":
    viewed = select(VideoView.video_id).where(VideoView.user_id == user_id)
    return (await session.execute(
        select(Video).where(
            Video.status == "approved",
            Video.content_type == "video",
            Video.uploader_user_id != user_id,
            ~Video.id.in_(viewed)
        ).order_by(func.random()).limit(1)
    )).scalar_one_or_none()

async def get_random_photo_for_user(
    session: AsyncSession,
    user_id: int
) -> "Video | None":
    return (await session.execute(
        select(Video).where(
            Video.status == "approved",
            Video.content_type == "photo",
            Video.uploader_user_id != user_id,
        ).order_by(func.random()).limit(1)
    )).scalar_one_or_none()

async def record_view_and_charge(
    session: AsyncSession,
    user_id: int,
    video_id: int
) -> bool:
    user = await get_user_by_id(session, user_id)
    if not user:
        return False
    cost = to_decimal(WATCH_COST)
    if user.balance < cost:
        return False
    await log_balance_change(
        session, user, -cost, "watch", source_id=video_id
    )
    user.balance -= cost
    from app.config import XP_PER_WATCH
    user.xp += XP_PER_WATCH
    view = VideoView(user_id=user_id, video_id=video_id)
    session.add(view)
    try:
        await session.commit()
        return True
    except Exception:
        await session.rollback()
        return False

```

```

async def record_photo_view(
    session: AsyncSession,
    user_id: int,
    photo_id: int
):
    view = VideoView(user_id=user_id, video_id=photo_id)
    session.add(view)
    try:
        await session.commit()
    except Exception:
        await session.rollback()

async def rate_video(
    session: AsyncSession,
    user_id: int,
    video_id: int,
    rating: int
):
    existing = (await session.execute(
        select(VideoRating).where(
            VideoRating.user_id == user_id,
            VideoRating.video_id == video_id
        )
    )).scalar_one_or_none()
    if existing:
        existing.rating = rating
    else:
        session.add(VideoRating(
            user_id=user_id,
            video_id=video_id,
            rating=rating
        ))
    await session.commit()

# =====
# ████████ / ████████
# =====
async def claim_daily_bonus(
    session: AsyncSession,
    user_id: int
) -> tuple[bool, object]:
    user = await get_user_by_id(session, user_id)
    if not user:
        return False, "██████████████████ ███ ████████"
    now = datetime.utcnow()
    if user.last_bonus_at and (now - user.last_bonus_at).total_seconds() < 86400:
        remaining = 86400 - (now - user.last_bonus_at).total_seconds()
        hours = int(remaining // 3600)
        minutes = int((remaining % 3600) // 60)
        return False, f"██ ████████ ██████████. ██████████ ████████ {hours}██ {minutes}██████"
    bonus = to_decimal(1.0)
    await log_balance_change(session, user, bonus, "daily_bonus")
    user.balance += bonus
    user.last_bonus_at = now
    await session.commit()
    return True, bonus

async def count_referrals(session: AsyncSession, user_id: int) -> int:
    return (await session.execute(
        select(func.count(User.id)).where(User.referred_by_user_id == user_id)
    )).scalar_one()

async def update_user_balance(
    session: AsyncSession,
    user_id: int,
    amount: Decimal,
    admin_id: int
) -> bool:
    user = await get_user_by_id(session, user_id)
    if not user:
        return False
    await log_balance_change(
        session, user, amount,
        "admin_balance",
        admin_id=admin_id,
        details=f"Manual by admin {admin_id}"
    )
    user.balance += amount
    await log_user_action(
        session, user_id,
        "balance_update",
        f"admin={admin_id}, amount={amount}"
    )

```

```

        await session.commit()
        return True

async def set_user_ban_status(
    session: AsyncSession,
    user_id: int,
    is_banned: bool,
    admin_id: int
) -> bool:
    user = await get_user_by_id(session, user_id)
    if not user:
        return False
    user.status = "banned" if is_banned else "active"
    await log_user_action(
        session, user_id,
        "ban_status_change",
        f"admin={admin_id}, banned={is_banned}"
    )
    await session.commit()
    return True

# =====
# ████████
# =====
async def create_payment(
    session: AsyncSession,
    user_id: int,
    pack_key: str
) -> "Payment | None":
    pack = STARS_PACKAGES.get(pack_key)
    if not pack:
        return None
    payload = f"{user_id}_{pack_key}_{uuid.uuid4().hex[:8]}"
    payment = Payment(
        user_id=user_id,
        payload=payload,
        stars_amount=pack["stars"],
        coins_amount=to_decimal(pack["coins"]),
        status="pending",
    )
    session.add(payment)
    await session.commit()
    return payment

async def create_custom_payment(
    session: AsyncSession,
    user_id: int,
    stars: int
) -> "Payment":
    coins = to_decimal(stars) * to_decimal(STARS_TO_COINS_RATE)
    payload = f"{user_id}_custom_{uuid.uuid4().hex[:8]}"
    payment = Payment(
        user_id=user_id,
        payload=payload,
        stars_amount=stars,
        coins_amount=coins,
        status="pending",
    )
    session.add(payment)
    await session.commit()
    return payment

async def apply_successful_payment(
    session: AsyncSession,
    payload: str
) -> "Payment | None":
    payment = (await session.execute(
        select(Payment).where(Payment.payload == payload)
    )).scalar_one_or_none()
    if not payment or payment.status == "completed":
        return None
    payment.status = "completed"
    user = await get_user_by_id(session, payment.user_id)
    if user:
        await log_balance_change(
            session, user, payment.coins_amount,
            "stars_payment",
            source_id=payment.id,
            details=f"stars={payment.stars_amount}, payload={payload}"
        )
        user.balance += payment.coins_amount
    await session.commit()
    return payment

```

```

# =====
# ████████
# =====
async def get_active_offers(session: AsyncSession) -> list["Offer"]:
    return (await session.execute(
        select(Offer).where(
            Offer.is_active == True,
            Offer.status == "approved"
        ).order_by(desc(Offer.created_at))
    )).scalars().all()

async def get_rentable_offers(session: AsyncSession) -> list["Offer"]:
    """████████ ██████████ ███ ████████."""
    return (await session.execute(
        select(Offer).where(
            Offer.is_active == True,
            Offer.status == "approved",
            Offer.is_rentable == True,
        ).order_by(desc(Offer.created_at))
    )).scalars().all()

async def get_offer_by_id(
    session: AsyncSession,
    offer_id: int
) -> "Offer | None":
    return (await session.execute(
        select(Offer).where(Offer.id == offer_id)
    )).scalar_one_or_none()

async def start_offer_participation(
    session: AsyncSession,
    user_id: int,
    offer_id: int
) -> tuple["OfferParticipation | None", bool]:
    existing = (await session.execute(
        select(OfferParticipation).where(
            OfferParticipation.user_id == user_id,
            OfferParticipation.offer_id == offer_id
        )
    )).scalar_one_or_none()
    if existing:
        return existing, False

    offer = await get_offer_by_id(session, offer_id)
    if not offer:
        return None, False

    user = await get_user_by_id(session, user_id)
    if user:
        await log_balance_change(
            session, user, offer.reward_preview,
            "offer_start", source_id=offer_id
        )
        user.balance += offer.reward_preview

    part = OfferParticipation(
        user_id=user_id,
        offer_id=offer_id,
        status="started",
        reward_given=offer.reward_preview,
    )
    session.add(part)
    await session.commit()
    return part, True

async def verify_offer_subscription(
    session: AsyncSession,
    user_id: int,
    offer_id: int
) -> bool:
    part = (await session.execute(
        select(OfferParticipation).where(
            OfferParticipation.user_id == user_id,
            OfferParticipation.offer_id == offer_id
        )
    )).scalar_one_or_none()
    if not part:
        return False
    if part.status == "completed":
        return True

    part.status = "completed"
    part.checked_at = datetime.utcnow()

```

[illegible]

```

"""
#####  #####  ##  #####  #####.
##### (rental, None) ##  #####  ##  (None, error_text).
"""
if rent_days < OFFER_MIN_RENT_DAYS or rent_days > OFFER_MAX_RENT_DAYS:
    return None, (
        f"#####  #####: "
        f"{OFFER_MIN_RENT_DAYS}-{OFFER_MAX_RENT_DAYS}."
    )

offer = await get_offer_by_id(session, offer_id)
if not offer:
    return None, "#####."
if not offer.is_rentable:
    return None, "#####."
if not offer.is_active:
    return None, "#####."

# #####  #####
active_count = (await session.execute(
    select(func.count(OfferRental.id)).where(
        OfferRental.offer_id == offer_id,
        OfferRental.status.in_(["active", "pending"])
    )
)).scalar_one()
if active_count >= offer.max_simultaneous_rentals:
    return None, (
        f"##### ({active_count}/{offer.max_simultaneous_rentals}).\n"
        f"#####."
    )

# #####  #####  #####
existing = (await session.execute(
    select(OfferRental).where(
        OfferRental.offer_id == offer_id,
        OfferRental.renter_user_id == user_id,
        OfferRental.status.in_(["pending", "active"])
    )
)).scalar_one_or_none()
if existing:
    return None, "#####."

cost = to_decimal(offer.rent_cost_per_day) * rent_days
user = await get_user_by_id(session, user_id)
if not user:
    return None, "#####."
if user.balance < cost:
    return None, f"#####.\n#####: {cost}, #####: {user.balance}"

await log_balance_change(
    session, user, -cost,
    "offer_rental", source_id=offer_id,
    details=f"##### {rent_days} #####: {channel_title}"
)
user.balance -= cost

rental = OfferRental(
    offer_id=offer_id,
    renter_user_id=user_id,
    renter_channel_title=channel_title,
    renter_channel_url=channel_url,
    rent_days=rent_days,
    cost_paid=cost,
    status="pending",
)
session.add(rental)
await session.commit()
return rental, None

async def get_user_rentals(
    session: AsyncSession,
    user_id: int
) -> list["OfferRental"]:
    return (await session.execute(
        select(OfferRental)
        .where(OfferRental.renter_user_id == user_id)
        .order_by(desc(OfferRental.created_at))
        .limit(10)
    )).scalars().all()

# =====
# #####
# =====
async def get_admin_extended_stats(session: AsyncSession) -> dict:
    users = (await session.execute(
        select(func.count(User.id))
    ))

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)).scalar_one()

vip = (await session.execute(
    select(func.count(User.id)).where(User.vip_until > datetime.utcnow())
)).scalar_one()

with_nickname = (await session.execute(
    select(func.count(User.id)).where(User.nickname_set == True)
)).scalar_one()

comments = (await session.execute(
    select(func.count(Comment.id))
)).scalar_one()

reactions = (await session.execute(
    select(func.count(ContentReaction.id))
)).scalar_one()

games = (await session.execute(
    select(func.count(GameHistory.id))
)).scalar_one()

offers = (await session.execute(
    select(func.count(Offer.id))
)).scalar_one()

total_balance = (await session.execute(
    select(func.sum(User.balance))
)).scalar_one() or Decimal("0")

total_admin_given = (await session.execute(
    select(func.sum(BalanceLog.amount)).where(
        BalanceLog.source == "admin_balance",
        BalanceLog.amount > 0
    )
)).scalar_one() or Decimal("0")

total_game_profit = (await session.execute(
    select(func.sum(GameHistory.result))
)).scalar_one() or Decimal("0")

try:
    active_rentals = (await session.execute(
        select(func.count(OfferRental.id)).where(
            OfferRental.status == "active"
        )
    )).scalar_one()

    total_rent_income = (await session.execute(
        select(func.sum(BalanceLog.amount)).where(
            BalanceLog.source == "offer_rental"
        )
    )).scalar_one() or Decimal("0")
except Exception:
    active_rentals = 0
    total_rent_income = Decimal("0")

return {
    "users": users,
    "vip": vip,
    "with_nickname": with_nickname,
    "comments": comments,
    "reactions": reactions,
    "games": games,
    "offers": offers,
    "active_rentals": active_rentals,
    "total_balance_in_system": total_balance,
    "total_admin_given": total_admin_given,
    "total_game_profit": total_game_profit,
    "total_rent_income": total_rent_income,
}

async def get_user_dossier(session: AsyncSession, user_id: int) -> dict | None:
    user = await get_user_by_id(session, user_id)
    if not user:
        return None

    games_count = (await session.execute(
        select(func.count(GameHistory.id)).where(GameHistory.user_id == user_id)
    )).scalar_one()

    game_profit = (await session.execute(
        select(func.sum(GameHistory.result)).where(GameHistory.user_id == user_id)
    )).scalar_one() or Decimal("0")

    suspicious_games = (await session.execute(
        select(GameHistory).where(

```

```

        GameHistory.user_id == user_id,
        GameHistory.result > to_decimal(50)
    ).order_by(desc(GameHistory.created_at))
)).scalars().all()

videos_uploaded = (await session.execute(
    select(func.count(Video.id)).where(Video.uploader_user_id == user_id)
)).scalar_one()

videos_watched = (await session.execute(
    select(func.count(VideoView.id)).where(VideoView.user_id == user_id)
)).scalar_one()

balance_logs = (await session.execute(
    select(BalanceLog).where(BalanceLog.user_id == user_id)
    .order_by(desc(BalanceLog.created_at))
    .limit(50)
)).scalars().all()

total_earned = (await session.execute(
    select(func.sum(BalanceLog.amount)).where(
        BalanceLog.user_id == user_id,
        BalanceLog.amount > 0
    )
)).scalar_one() or Decimal("0")

total_spent = (await session.execute(
    select(func.sum(BalanceLog.amount)).where(
        BalanceLog.user_id == user_id,
        BalanceLog.amount < 0
    )
)).scalar_one() or Decimal("0")

admin_given = (await session.execute(
    select(func.sum(BalanceLog.amount)).where(
        BalanceLog.user_id == user_id,
        BalanceLog.source == "admin_balance",
        BalanceLog.amount > 0
    )
)).scalar_one() or Decimal("0")

logs = (await session.execute(
    select(UserActionLog).where(UserActionLog.user_id == user_id)
    .order_by(desc(UserActionLog.created_at))
    .limit(20)
)).scalars().all()

return {
    "user": user,
    "games_count": games_count,
    "game_profit": game_profit,
    "suspicious_games": suspicious_games,
    "videos_uploaded": videos_uploaded,
    "videos_watched": videos_watched,
    "balance_logs": balance_logs,
    "total_earned": total_earned,
    "total_spent": total_spent,
    "admin_given": admin_given,
    "logs": logs,
}

```

```

=====
FILE: app\user_handlers.py
=====

```

```

import re
import uuid
from datetime import datetime, timedelta
from decimal import Decimal

from aiogram import Router, F
from aiogram.filters import CommandStart, CommandObject
from aiogram.fsm.context import FSMContext
from aiogram.fsm.state import State, StatesGroup
from aiogram.types import (
    Message, CallbackQuery,
    LabeledPrice, PreCheckoutQuery,
    InlineKeyboardMarkup, InlineKeyboardButton,
)
from sqlalchemy import select, func, desc

from app.config import (
    ADMINS, WATCH_COST, STARS_PACKAGES, STARS_TO_COINS_RATE,
    XP_PER_WATCH, XP_PER_UPLOAD, XP_PER_RATING,
    XP_PER_COMMENT, XP_PER_REACTION, XP_PER_GAME,
    PIN_OFFER_COST, VIP_PRICE_STARS, VIP_DURATION_DAYS, VIP_BONUS_MULTIPLIER,
    LEVEL_XP_BASE, LEVEL_XP_MULTIPLIER,
    DAILY_QUESTS, PREMIUM_DAILY_QUESTS,
    COMMENTS_PER_10_MIN,

```

```

NICKNAME_CHANGE_COST, NICKNAME_MIN_LENGTH, NICKNAME_MAX_LENGTH,
OFFER_DEFAULT_RENT_COST_PER_DAY, OFFER_MIN_RENT_DAYS, OFFER_MAX_RENT_DAYS,
REFERRAL_REWARD_INVITER, REFERRAL_REWARD_NEW_USER,
)
from app.db import async_session
from app.models import (
    User, Video, VideoView, Comment, ContentReaction,
    DailyQuestProgress, GameHistory, OfferRental, Offer,
)
from app.services import (
    get_or_create_user, get_user, get_user_by_id,
    save_video, save_photo,
    get_random_video_for_user, get_random_photo_for_user,
    record_view_and_charge, record_photo_view,
    rate_video, claim_daily_bonus, count_referrals,
    create_payment, create_custom_payment, apply_successful_payment,
    get_active_offers, get_rentable_offers, get_offer_by_id,
    start_offer_participation, verify_offer_subscription,
    create_offer_rental, get_user_rentals,
    log_user_action, to_decimal,
    set_display_name, get_display_name, log_balance_change,
)
from app.keyboards import (
    rules_keyboard, main_menu,
    video_rating_keyboard, photo_actions_keyboard,
    watch_choice_keyboard, buy_coins_keyboard, vip_buy_keyboard,
    offers_list_keyboard, offer_view_keyboard,
    offer_rent_keyboard, rent_days_keyboard,
    games_menu_keyboard, tops_menu_keyboard,
    quests_keyboard, reaction_menu_keyboard,
    BTN_WATCH, BTN_UPLOAD, BTN_PROFILE, BTN_BUY,
    BTN_OFFERS, BTN_REFERRALS, BTN_BONUS, BTN_ADMIN,
    BTN_GAMES, BTN_TOPS, BTN_QUESTS, BTN_VIP, BTN_LEVEL,
)
from app.logger import get_logger

logger = get_logger(__name__)
router = Router()

# =====
# STATES
# =====
class NicknameState(StatesGroup):
    waiting_nickname = State()

class CommentState(StatesGroup):
    waiting_text = State()

class CustomBuyState(StatesGroup):
    waiting_stars = State()

class RentOfferState(StatesGroup):
    waiting_channel_title = State()
    waiting_channel_url = State()
    waiting_days = State()

# =====
# HELPERS
# =====
def is_any_admin(telegram_id: int, user_obj=None) -> bool:
    if telegram_id in ADMINS:
        return True
    if user_obj and user_obj.is_admin:
        return True
    return False

def calc_level_xp_required(level: int) -> int:
    return int(LEVEL_XP_BASE * (LEVEL_XP_MULTIPLIER ** (level - 1)))

def calc_level_from_xp(xp: int) -> int:
    level = 1
    remaining = xp
    while True:
        required = calc_level_xp_required(level)
        if remaining < required:
            break
        remaining -= required
        level += 1
    return level

```

```

def is_vip(user) -> bool:
    return bool(user.vip_until and user.vip_until > datetime.utcnow())

async def require_nickname(message: Message, user) -> bool:
    """
    """
    if user.nickname_set and user.display_name:
        return True
    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■",
            callback_data="set_nickname_start"
        )]
    ])
    await message.answer(
        f"■ <b>■!</b>\n\n"
        f"• {NICKNAME_MIN_LENGTH} {NICKNAME_MAX_LENGTH} \n"
        f"• (■/■), ■, _ ■ -\n"
        f"• \n\n"
        f"■ - ■!\n"
        f"■ {NICKNAME_CHANGE_COST} ■.",
        parse_mode="HTML",
        reply_markup=kb
    )
    return False

async def _level_up_check(session, user, message_or_callback):
    """
    """
    new_level = calc_level_from_xp(user.xp)
    if new_level > user.level:
        user.level = new_level
        await session.commit()
        if hasattr(message_or_callback, "answer"):
            await message_or_callback.answer(
                f"■! ■ <b>{new_level}</b>!",
                parse_mode="HTML"
            )
        else:
            await message_or_callback.message.answer(
                f"■! ■ <b>{new_level}</b>!",
                parse_mode="HTML"
            )

# =====
# NICKNAME FLOW
# =====
@router.callback_query(F.data == "set_nickname_start")
async def set_nickname_start(callback: CallbackQuery, state: FSMContext):
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        is_first = not user.nickname_set
        cost_text = "■" if is_first else f"{NICKNAME_CHANGE_COST} ■"

    await state.set_state(NicknameState.waiting_nickname)
    await callback.message.answer(
        f"=■ ({cost_text}):\n\n"
        f"• {NICKNAME_MIN_LENGTH} {NICKNAME_MAX_LENGTH} \n"
        f"• (■/■), ■, _ ■ -\n"
        f"• ■, ■, ■"
    )
    await callback.answer()

@router.message(NicknameState.waiting_nickname)
async def process_nickname(message: Message, state: FSMContext):
    name = (message.text or "").strip()
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            await state.clear()
            return
        ok, msg = await set_display_name(session, user, name)

    await message.answer(msg, parse_mode="HTML")
    if ok:
        await state.clear()
        async with async_session() as session:
            user = await get_user(session, message.from_user.id)
            admin_flag = is_admin(message.from_user.id, user)
            await message.answer(
                "■!",
                reply_markup=main_menu(is_admin=admin_flag)
            )

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# =====  
# START / RULES  
# =====  
  
@router.message(CommandStart())  
async def cmd_start(message: Message, command: CommandObject, state: FSMContext):  
    if not message.from_user:  
        return  
    await state.clear()  
    referral_code = (command.args or "").strip() or None  
  
    async with async_session() as session:  
        user, is_new = await get_or_create_user(  
            session,  
            message.from_user.id,  
            message.from_user.username,  
            message.from_user.first_name,  
            message.from_user.last_name,  
            referral_code,  
        )  
        if user.status == "banned":  
            await message.answer("❗ ❗ 🚫🚫🚫🚫🚫 🚫 🚫🚫🚫.")  
            return  
  
        if not user.agreed_to_rules:  
            await message.answer(  
                "❗ <b>👉🏻📖📖📖📖📖</b>\n\n"  
                "1. 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 (👤 👤 👤-👤  
👤). \n\n"  
                "2. 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤.\n\n"  
                "3. 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤.  
\n\n"  
                "👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤.",  
                parse_mode="HTML",  
                reply_markup=rules_keyboard()  
            )  
            return  
  
        if not user.nickname_set or not user.display_name:  
            kb = InlineKeyboardMarkup(inline_keyboard=[  
                [InlineKeyboardButton(  
                    text="➡️ 👤 👤 👤 👤 👤 👤 👤 👤",  
                    callback_data="set_nickname_start"  
                )]  
            ])  
            await message.answer(  
                "❗ 👤 👤 👤 👤 👤 👤! \n\n"  
                "👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤.\n\n"  
                f"👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤!\n\n"  
                f"• 👤 {NICKNAME_MIN_LENGTH} 👤 {NICKNAME_MAX_LENGTH} 👤 👤 👤 👤\n\n"  
                f"• 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 - ",  
                parse_mode="HTML",  
                reply_markup=kb  
            )  
            return  
  
            admin_flag = is_admin(message.from_user.id, user)  
            vip_str = " 👤" if is_vip(user) else ""  
            await message.answer(  
                f"❗ 👤 👤 👤 👤, <b>{get_display_name(user)}</b>{vip_str}! \n\n"  
                f"❗ 👤 👤 👤 👤: <b>{user.balance}</b> 👤 👤 👤 👤",  
                parse_mode="HTML",  
                reply_markup=main_menu(is_admin=admin_flag)  
            )  
  
@router.callback_query(F.data == "accept_rules")  
async def accept_rules(callback: CallbackQuery):  
    async with async_session() as session:  
        user = await get_user(session, callback.from_user.id)  
        if not user:  
            await callback.answer()  
            return  
        user.agreed_to_rules = True  
        await session.commit()  
  
        kb = InlineKeyboardMarkup(inline_keyboard=[  
            [InlineKeyboardButton(  
                text="➡️ 👤 👤 👤 👤 👤 👤 👤 👤",  
                callback_data="set_nickname_start"  
            )]  
        ])  
        await callback.message.answer(  
            "❗ 👤 👤 👤 👤 👤 👤! \n\n"  
            "❗ 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤.\n\n"  
            f"• 👤 {NICKNAME_MIN_LENGTH} 👤 {NICKNAME_MAX_LENGTH} 👤 👤 👤 👤\n\n"  
            f"• 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 👤 (👤/👤), 👤 👤 👤 👤, - ",
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        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

# =====
# ADMIN REDIRECT
# =====
@router.message(F.text == BTN_ADMIN)
async def cmd_admin_redirect(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if is_admin(message.from_user.id, user):
            from app.admin_handlers import cmd_admin
            await cmd_admin(message)

# =====
# PROFILE
# =====
@router.message(F.text == BTN_PROFILE)
async def show_profile(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return

        refs = await count_referrals(session, user.id)
        vip_str = ""
        if is_vip(user):
            vip_str = f"\n■ VIP ■■■: {user.vip_until.strftime('%d.%m.%Y')}"

        level = user.level
        xp_spent = sum(calc_level_xp_required(l) for l in range(1, level))
        xp_current = user.xp - xp_spent
        xp_needed = calc_level_xp_required(level)
        progress = max(0, min(10, int((xp_current / max(xp_needed, 1)) * 10)))
        bar = "■" * progress + "■" * (10 - progress)

        kb = InlineKeyboardMarkup(inline_keyboard=[
            [InlineKeyboardButton(
                text="=■ ■■■■■■■■ ■■■",
                callback_data="set_nickname_start"
            )]
        ])
        text = (
            f"■ <b>■■■■■■■■</b>\n\n"
            f"■ ■■■: <b>{get_display_name(user)}</b>\n"
            f"■ TG ID: <code>{user.telegram_id}</code>\n"
            f"■ ■■■■■■■■: <b>{user.balance}</b> ■■■■■\n"
            f"■ ■■■■■■■■: <b>{user.level}</b>\n"
            f"■ XP: {xp_current}/{xp_needed} [{bar}]\n"
            f"■ ■■■■■■■■: {refs}\n"
            f"■ ■■■. ■■■■■■■■■■: {user.referral_earnings} ■■■■■\n"
            f"■ ■■■■■: {user.status}"
            f"{vip_str}\n\n"
            f"■ ■■■■■ ■■■: {NICKNAME_CHANGE_COST} ■■■■■"
        )
    await message.answer(text, parse_mode="HTML", reply_markup=kb)
    await log_user_action(session, user.id, "view_profile")

# =====
# LEVEL
# =====
@router.message(F.text == BTN_LEVEL)
async def show_level(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return

        level = user.level
        xp_spent = sum(calc_level_xp_required(l) for l in range(1, level))
        xp_current = user.xp - xp_spent
        xp_needed = calc_level_xp_required(level)
        progress = max(0, min(10, int((xp_current / max(xp_needed, 1)) * 10)))
        bar = "■" * progress + "■" * (10 - progress)

        text = (
            f"■ <b>■■■■■■■■: {level}</b>\n\n"
            f"XP: {xp_current}/{xp_needed}\n"
            f"[{bar}]\n\n"

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f"■ ■■■ ■■■■■■■■ XP:\n"
f"• ■■■■■■■■ ■■■■■■■■: +{XP_PER_WATCH} XP\n"
f"• ■■■■■■■■ ■■■■■■■■: +{XP_PER_UPLOAD} XP\n"
f"• ■■■■■■■■ ■■■■■■■■: +{XP_PER_RATING} XP\n"
f"• ■■■■■■■■■■■■: +{XP_PER_COMMENT} XP\n"
f"• ■■■■■■■■: +{XP_PER_REACTION} XP\n"
f"• ■■■■■: +{XP_PER_GAME} XP"
)
await message.answer(text, parse_mode="HTML")

# =====
# VIP
# =====
@router.message(F.text == BTN_VIP)
async def show_vip(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return

    if is_vip(user):
        await message.answer(
            f"■ <b>■ VIP!</b>\n\n"
            f"■: <b>{user.vip_until.strftime('%d.%m.%Y %H:%M')}</b>\n\n"
            f"■■■■■■■■: \n"
            f"• ■■■■■■■■ ■■■■■ x{VIP_BONUS_MULTIPLIER}\n"
            f"• ■■■■■ 50% ■■ ■■■■■■■■\n"
            f"• VIP ■■■■■■■■",
            parse_mode="HTML"
        )
    else:
        await message.answer(
            f"■ <b>VIP ■■■■■■■■</b>\n\n"
            f"■■■■■■■■■■: <b>{VIP_PRICE_STARS} Stars</b>\n"
            f"■■■■■■■■■■: {VIP_DURATION_DAYS} ■■■■\n\n"
            f"■■■■■■■■■■: \n"
            f"• ■■■■■■■■ ■■■■■ x{VIP_BONUS_MULTIPLIER}\n"
            f"• ■■■■■ 50% ■■ ■■■■■■■■\n"
            f"• VIP ■■■■■■■■",
            parse_mode="HTML",
            reply_markup=vip_buy_keyboard()
        )

@router.callback_query(F.data == "buy_vip")
async def buy_vip(callback: CallbackQuery):
    await callback.message.answer_invoice(
        title="VIP ■■■■■■■■",
        description=f"VIP ■■ {VIP_DURATION_DAYS} ■■■■",
        payload=f"vip_{callback.from_user.id}_{uuid.uuid4().hex[:6]}",
        currency="XTR",
        prices=[LabeledPrice(label="VIP", amount=VIP_PRICE_STARS)]
    )
    await callback.answer()

# =====
# WATCH
# =====
@router.message(F.text == BTN_WATCH)
async def btn_watch(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if user.status == "banned":
            await message.answer("■ ■■ ■■■■■■■■■■■■■■.")
            return
        if not await require_nickname(message, user):
            return
    await message.answer("■■■ ■■■■■■■■?", reply_markup=watch_choice_keyboard())

@router.callback_query(F.data == "watch_video_content")
async def watch_video_content(callback: CallbackQuery):
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return

    cost = to_decimal(WATCH_COST)
    if is_vip(user):
        cost = round(cost * to_decimal(0.5), 2)
```

[illegible]

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        user.xp += XP_PER_RATING
        await _level_up_check(session, user, callback)
        await _update_quest_progress(session, user.id, "rate", 1)

    await callback.answer(f"■ ■■■■■■ {rating} ■■■■■■!")

# =====
# COMMENTS
# =====
@router.callback_query(F.data.startswith("comments:"))
async def cb_comments(callback: CallbackQuery):
    video_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        comments = (await session.execute(
            select(Comment)
            .where(Comment.video_id == video_id)
            .order_by(desc(Comment.created_at))
            .limit(10)
        )).scalars().all()

    text = f"■ <b>■■■■■■■■■■ ■ ■■■■■ ■{video_id}</b>\n\n"
    if not comments:
        text += "■■■■■■■■■■ ■■■■ ■■■. ■■■■■ ■■■■■!"
    else:
        for c in comments:
            u = await get_user_by_id(session, c.user_id)
            name = get_display_name(u) if u else "???"
            text += f"■ <b>{name}</b>: {c.text}\n"

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ■■■■■■■■",
            callback_data=f"add_comment:{video_id}"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■",
            callback_data=f"reactions:{video_id}"
        )],
    ])
    await callback.message.answer(text, parse_mode="HTML", reply_markup=kb)
    await callback.answer()

@router.callback_query(F.data.startswith("add_comment:"))
async def add_comment_start(callback: CallbackQuery, state: FSMContext):
    video_id = int(callback.data.split(":")[1])
    await state.set_state(CommentState.waiting_text)
    await state.update_data(video_id=video_id)
    await callback.message.answer("■ ■■■■■■■■ ■■■■■■■■■■:")
    await callback.answer()

@router.message(CommentState.waiting_text)
async def process_comment(message: Message, state: FSMContext):
    data = await state.get_data()
    video_id = data.get("video_id")
    if not video_id:
        await state.clear()
        return

    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            await state.clear()
            return

    # ■■■■■■■■
    ten_min_ago = datetime.utcnow() - timedelta(minutes=10)
    recent = (await session.execute(
        select(func.count(Comment.id)).where(
            Comment.user_id == user.id,
            Comment.created_at >= ten_min_ago
        )
    )).scalar_one()
    if recent >= COMMENTS_PER_10_MIN:
        await message.answer(
            f"■ ■ ■■■■■ {COMMENTS_PER_10_MIN} ■■■■■■■■■■ ■■ 10 ■■■■■."
        )
        await state.clear()
        return

    from app.models import Comment as CommentModel
    session.add(CommentModel(
        user_id=user.id,
        video_id=video_id,
        text=message.text
    ))

```

[illegible]

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        if not user.nickname_set:
            await require_nickname(message, user)
            return

        v = message.video
        saved = await save_video(
            session, user.id,
            v.file_id, v.file_unique_id,
            v.duration, v.file_size
        )
        user.xp += XP_PER_UPLOAD
        await _level_up_check(session, user, message)
        await _update_quest_progress(session, user.id, "upload", 1)

    await message.answer(
        f"■ ■■■■■ ■#{saved.id} ■■■■■■■■■■ ■■ ■■■■■■■■■■!"
    )

@router.message(F.photo)
async def handle_photo_upload(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user or user.status == "banned":
            return
        if not user.agreed_to_rules:
            await message.answer("■■■■■■■■ ■■■■■■■ ■■■■■■■ /start")
            return
        if not user.nickname_set:
            await require_nickname(message, user)
            return

        p = message.photo[-1]
        saved = await save_photo(
            session, user.id,
            p.file_id, p.file_unique_id,
            p.file_size
        )
        user.xp += XP_PER_UPLOAD
        await _level_up_check(session, user, message)
        await _update_quest_progress(session, user.id, "upload", 1)

    await message.answer(
        f"■ ■■■■ ■#{saved.id} ■■■■■■■■■■ ■■ ■■■■■■■■■■!"
    )

# =====
# BONUS
# =====
@router.message(F.text == BTN_BONUS)
async def btn_bonus(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return
        ok, result = await claim_daily_bonus(session, user.id)
    if ok:
        await message.answer(
            f"■ ■■■■■ ■■■■■■■: <b>+{result} ■■■■■</b>!",
            parse_mode="HTML"
        )
    else:
        await message.answer(str(result))

# =====
# REFERRALS
# =====
@router.message(F.text == BTN_REFERRALS)
async def btn_referrals(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return
        refs = await count_referrals(session, user.id)

    bot_info = await message.bot.get_me()
    ref_link = f"https://t.me/{bot_info.username}?start={user.referral_code}"
    await message.answer(
        f"■ <b>■■■■■■■■■■</b>\n\n"
        f"■■■■ ■■■■■: \n<code>{ref_link}</code>\n\n"
        f"■■■■■■■■■■: <b>{refs}</b>\n\n"
        f"■■■■■■■■■■: <b>{user.referral_earnings}</b> ■■■■■\n\n"
    )

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```
f"[REDACTED]:\n"  
f". [REDACTED] +{REFERRAL_REWARD_INVITER} [REDACTED]\n"  
f". [REDACTED] +{REFERRAL_REWARD_NEW_USER} [REDACTED]",  
parse_mode="HTML")  
  
# =====  
# BUY COINS  
# =====  
@router.message(F.text == BTN_BUY)  
async def btn_buy(message: Message):  
    async with async_session() as session:  
        user = await get_user(session, message.from_user.id)  
        if not user:  
            return  
        if not await require_nickname(message, user):  
            return  
    await message.answer(  
        "[REDACTED]<b>[REDACTED]</b>\n\n[REDACTED]:",  
        parse_mode="HTML",  
        reply_markup=buy_coins_keyboard()  
    )  
  
@router.callback_query(F.data.startswith("buy:"))  
async def cb_buy_pack(callback: CallbackQuery):  
    pack_key = callback.data.split(":")[1]  
    pack = STARS_PACKAGES.get(pack_key)  
    if not pack:  
        await callback.answer("[REDACTED].", show_alert=True)  
        return  
  
    async with async_session() as session:  
        user = await get_user(session, callback.from_user.id)  
        if not user:  
            await callback.answer()  
            return  
        payment = await create_payment(session, user.id, pack_key)  
  
    await callback.message.answer_invoice(  
        title=f"[REDACTED] {pack['title']}",  
        description=f"{pack['coins']} [REDACTED] Stars",  
        payload=payment.payload,  
        currency="XTR",  
        prices=[LabeledPrice(label=pack['title'], amount=pack['stars'])]  
    )  
    await callback.answer()  
  
@router.callback_query(F.data == "buy_custom")  
async def cb_buy_custom(callback: CallbackQuery, state: FSMContext):  
    await state.set_state(CustomBuyState.waiting_stars)  
    await callback.message.answer("[REDACTED] Stars ([REDACTED]. 1):")  
    await callback.answer()  
  
@router.message(CustomBuyState.waiting_stars)  
async def process_custom_stars(message: Message, state: FSMContext):  
    if not message.text or not message.text.isdigit():  
        await message.answer("[REDACTED].")  
        return  
    stars = int(message.text)  
    if stars < 1:  
        await message.answer("[REDACTED] 1 Star.")  
        return  
  
    async with async_session() as session:  
        user = await get_user(session, message.from_user.id)  
        if not user:  
            await state.clear()  
            return  
        payment = await create_custom_payment(session, user.id, stars)  
        coins = int(stars * STARS_TO_COINS_RATE)  
  
    await message.answer_invoice(  
        title=f"[REDACTED] {coins} [REDACTED]",  
        description=f"{coins} [REDACTED] Stars",  
        payload=payment.payload,  
        currency="XTR",  
        prices=[LabeledPrice(label=f"{coins} [REDACTED]", amount=stars)]  
    )  
    await state.clear()  
  
@router.pre_checkout_query()  
async def pre_checkout(query: PreCheckoutQuery):  
    await query.answer(ok=True)
```

```
@router.message(F.successful_payment)
async def successful_payment(message: Message):
    payload = message.successful_payment.invoice_payload

    if payload.startswith("vip_"):
        async with async_session() as session:
            user = await get_user(session, message.from_user.id)
            if user:
                now = datetime.utcnow()
                user.vip_until = (
                    user.vip_until + timedelta(days=VIP_DURATION_DAYS)
                    if user.vip_until and user.vip_until > now
                    else now + timedelta(days=VIP_DURATION_DAYS)
                )
                await log_user_action(
                    session, user.id,
                    "buy_vip",
                    f"until={user.vip_until}"
                )
                await session.commit()
            await message.answer(
                f"■ VIP ██████████ {VIP_DURATION_DAYS} ██████!"
            )
    else:
        async with async_session() as session:
            payment = await apply_successful_payment(session, payload)
            if payment:
                await message.answer(
                    f"■ ██████████ ██████████!\n"
                    f"■ ██████████: <b>{payment.coins_amount}</b> ██████",
                    parse_mode="HTML"
                )
            else:
                await message.answer("■ ██████████ ██████████!")

# =====
# OFFERS
# =====
@router.message(F.text == BTN_OFFERS)
async def btn_offers(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return
        offers = await get_active_offers(session)

    if not offers:
        await message.answer("■ ██████████ ██████████ ██████████. ██████████ ██████████!")
        return

    kb = InlineKeyboardMarkup(inline_keyboard=[
        [InlineKeyboardButton(
            text="■ ██████████ (████████)",
            callback_data="offers_participation"
        )],
        [InlineKeyboardButton(
            text="■ ██████████ ██████████ ██████████",
            callback_data="offers_rent_list"
        )],
        [InlineKeyboardButton(
            text="■ ███ ██████████",
            callback_data="my_rentals"
        )],
    ])
    await message.answer(
        "■ <b>████████</b></b>\n\n████████████████████ ██████████:",
        parse_mode="HTML",
        reply_markup=kb
    )

@router.callback_query(F.data == "offers_participation")
async def offers_participation(callback: CallbackQuery):
    async with async_session() as session:
        offers = await get_active_offers(session)

    if not offers:
        await callback.message.answer("■ ██████████ ██████████ ██████████.")
        await callback.answer()
        return

    await callback.message.answer(
        "■ <b>████████████████████ ██████████</b>\n\n████████████████████ ██████████:",

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        parse_mode="HTML",
        reply_markup=offers_list_keyboard(offers)
    )
    await callback.answer()

@router.callback_query(F.data == "back_to_offers")
async def back_to_offers(callback: CallbackQuery):
    await offers_participation(callback)

@router.callback_query(F.data.startswith("offer_open:"))
async def cb_offer_open(callback: CallbackQuery):
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        offer = await get_offer_by_id(session, offer_id)
        if not offer:
            await callback.answer("❌ Offer not found.", show_alert=True)
            return

        # 📄-📄 📄📄📄📄📄📄
        from sqlalchemy import select as sa_select
        from app.models import OfferParticipation
        participants = (await session.execute(
            sa_select(func.count(OfferParticipation.id)).where(
                OfferParticipation.offer_id == offer_id
            )
        )).scalar_one()

    text = (
        f"📄 <b>{offer.title}</b>\n\n"
        f"{offer.description}\n\n"
        f"📄 📄📄📄📄📄📄: <b>{offer.reward_preview}</b> 📄📄📄\n"
        f"📄 📄📄📄 📄📄📄📄📄📄: <b>{offer.reward_final}</b> 📄📄📄\n"
        f"📄 📄📄📄📄📄: {participants}"
    )

    # 📄📄 📄📄📄 📄📄📄📄📄📄📄📄 - 📄📄📄📄📄 📄📄📄📄
    kb_rows = [
        [InlineKeyboardButton(
            text="📄 📄📄📄📄 📄 📄📄📄",
            url=offer.channel_url
        )],
        [InlineKeyboardButton(
            text="📄 📄📄📄📄📄📄",
            callback_data=f"offer_start:{offer_id}"
        )],
        [InlineKeyboardButton(
            text="📄 📄📄📄📄📄 📄📄📄📄",
            callback_data=f"offer_check:{offer_id}"
        )],
    ]
    if getattr(offer, "is_rentable", False):
        kb_rows.append([InlineKeyboardButton(
            text="📄 📄📄📄📄 📄📄",
            callback_data=f"rent_offer:{offer_id}"
        )])
    kb_rows.append([InlineKeyboardButton(
        text="📄 📄📄📄",
        callback_data="offers_participation"
    )])

    await callback.message.answer(
        text, parse_mode="HTML",
        reply_markup=InlineKeyboardMarkup(inline_keyboard=kb_rows)
    )
    await callback.answer()

@router.callback_query(F.data.startswith("offer_start:"))
async def cb_offer_start(callback: CallbackQuery):
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        part, is_new = await start_offer_participation(session, user.id, offer_id)
        if part is None:
            await callback.answer("❌ Offer not found.", show_alert=True)
            return
        if not is_new:
            await callback.answer("📄 📄📄 📄📄📄📄📄!", show_alert=True)
            return
        offer = await get_offer_by_id(session, offer_id)

    await callback.answer(
        f"📄 📄📄📄📄 {offer.reward_preview} 📄📄📄!\n"
    )

```

```
f"██████████ ██████████ «██████████».",
show_alert=True
)

@router.callback_query(F.data.startswith("offer_check:"))
async def cb_offer_check(callback: CallbackQuery):
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        result = await verify_offer_subscription(session, user.id, offer_id)
        if result:
            offer = await get_offer_by_id(session, offer_id)
            await callback.answer(
                f"██████████! ██████████ {offer.reward_final} ██████████!",
                show_alert=True
            )
        else:
            await callback.answer(
                "██████████ ██████████ ██████████.",
                show_alert=True
            )

# =====
# ██████████ ██████████ ██████████
# =====
@router.callback_query(F.data == "offers_rent_list")
async def offers_rent_list(callback: CallbackQuery):
    async with async_session() as session:
        offers = await get_rentable_offers(session)

if not offers:
    await callback.message.answer(
        "██████████ ██████████ ██████████."
    )
    await callback.answer()
    return

text = "██████████ <b>██████████ ██████████ ███████████</b>\n\n"
text += (
    "██████████ ██████████ ██████████ ██████████ ██████████!\n"
    "██████████ ██████████ ██████████ ██████████ ██████████.\n\n"
    "██████████ ██████████:"
)
kb_buttons = []
for o in offers:
    active_count = 0
    async with async_session() as session:
        try:
            active_count = (await session.execute(
                select(func.count(OfferRental.id)).where(
                    OfferRental.offer_id == o.id,
                    OfferRental.status.in_(["active", "pending"])
                )
            ).scalar_one())
        except Exception:
            pass
    slots_left = o.max_simultaneous_rentals - active_count
    kb_buttons.append([InlineKeyboardButton(
        text=(
            f"██████████ {o.title[:30]} | "
            f"{o.rent_cost_per_day} ██████████/██████████ | "
            f"██████████: {slots_left}/{o.max_simultaneous_rentals}"
        ),
        callback_data=f"rent_offer:{o.id}"
    )])

kb_buttons.append([InlineKeyboardButton(
    text="██████████",
    callback_data="btn_offers_back"
)])

await callback.message.answer(
    text, parse_mode="HTML",
    reply_markup=InlineKeyboardMarkup(inline_keyboard=kb_buttons)
)
await callback.answer()

@router.callback_query(F.data == "btn_offers_back")
async def btn_offers_back(callback: CallbackQuery):
    async with async_session() as session:
        offers = await get_active_offers(session)
    kb = InlineKeyboardMarkup(inline_keyboard=[
```

```

        [InlineKeyboardButton(
            text="■ ■■■■■■ (■■■■■■)",
            callback_data="offers_participation"
        )],
        [InlineKeyboardButton(
            text="■ ■■■■■■■■■■ ■■■■■■■■ ■■■■",
            callback_data="offers_rent_list"
        )],
        [InlineKeyboardButton(
            text="■ ■■■ ■■■■■■",
            callback_data="my_rentals"
        )],
    ])
    await callback.message.answer(
        "■ <b>■■■■■■</b></b>\n\n■■■■■■■ ■■■■■:",
        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

@router.callback_query(F.data.startswith("rent_offer:"))
async def rent_offer_start(callback: CallbackQuery, state: FSMContext):
    offer_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        offer = await get_offer_by_id(session, offer_id)
        if not offer or not offer.is_rentable:
            await callback.answer("■■■■■■ ■■■■■■■■■■.", show_alert=True)
            return

        # ■■■■■■■■ ■■■■
        active_count = (await session.execute(
            select(func.count(OfferRental.id)).where(
                OfferRental.offer_id == offer_id,
                OfferRental.status.in_(["active", "pending"])
            )
        )).scalar_one()
        slots_left = offer.max_simultaneous_rentals - active_count

    if slots_left <= 0:
        await callback.answer(
            "■ ■■■ ■■■■ ■■■■. ■■■■■■■■ ■■■■.",
            show_alert=True
        )
        return

    await state.set_state(RentOfferState.waiting_channel_title)
    await state.update_data(offer_id=offer_id)
    await callback.message.answer(
        f"■ <b>■■■■■■ ■■■■ ■: {offer.title}</b>\n\n"
        f"■ ■■■■■■■■■: {offer.rent_cost_per_day} ■■■■/■■■■\n"
        f"■ ■■■■■■■■■ ■■■■■: {slots_left}/{offer.max_simultaneous_rentals}\n\n"
        f"■ 1/3: ■■■■■■ ■■■■■■ ■■■■■ ■■■■■:",
        parse_mode="HTML"
    )
    await callback.answer()

@router.message(RentOfferState.waiting_channel_title)
async def rent_channel_title(message: Message, state: FSMContext):
    title = (message.text or "").strip()
    if len(title) < 2 or len(title) > 100:
        await message.answer("■ ■■■■■■■■ ■■ 2 ■■ 100 ■■■■■■■■.")
        return
    await state.update_data(channel_title=title)
    await state.set_state(RentOfferState.waiting_channel_url)
    await message.answer(
        "■■■ 2/3: ■■■■■■ ■■■■■ ■■ ■■ ■■■■ (https://t.me/...):"
    )

@router.message(RentOfferState.waiting_channel_url)
async def rent_channel_url(message: Message, state: FSMContext):
    url = (message.text or "").strip()
    if not (url.startswith("https://t.me/") or url.startswith("t.me/")):
        await message.answer(
            "■ ■■■■■■ ■■■■■ ■■■■■■■■■ ■ https://t.me/ ■■■ t.me/"
        )
        return
    await state.update_data(channel_url=url)
    await state.set_state(RentOfferState.waiting_days)

    data = await state.get_data()
    offer_id = data.get("offer_id")

    await message.answer(
        f"■■■ 3/3: ■■■■■■■■ ■■■■■■■■■ ■■■■ ■■■■■\n"
        f"({{OFFER_MIN_RENT_DAYS}} ■■ {{OFFER_MAX_RENT_DAYS}}):",

```

[illegible]

[illegible]

```

        reply_markup=kb
    )
    await callback.answer()

@router.callback_query(F.data.startswith("dice_bet:"))
async def dice_bet(callback: CallbackQuery):
    bet = int(callback.data.split(":")[1])
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        if user.balance < to_decimal(bet):
            await callback.answer("❌ insufficient balance!", show_alert=True)
            return

        dice_msg = await callback.message.answer_dice(emoji="🎲")
        dice_value = dice_msg.dice.value

        user.balance -= to_decimal(bet)

        if dice_value >= 4:
            win = to_decimal(bet) * 2
            user.balance += win
            net = win - to_decimal(bet)
            result_text = f"🎲 {dice_value}: {dice_value}\n🎲 ! +{win} 🎲"
        else:
            net = -to_decimal(bet)
            result_text = f"🎲 {dice_value}: {dice_value}\n🎲 -{bet} 🎲"

        user.xp += XP_PER_GAME
        await _level_up_check(session, user, callback)

        session.add(GameHistory(
            user_id=user.id,
            game_type="dice",
            bet=to_decimal(bet),
            result=net,
            details=f"dice={dice_value}"
        ))
        await log_balance_change(
            session, user, net, "game_dice",
            details=f"bet={bet}, dice={dice_value}"
        )
        await session.commit()

    await callback.message.answer(
        f"{result_text}\n🎲 {user.balance}"
    )
    await callback.answer()

@router.callback_query(F.data == "game_coinflip")
async def game_coinflip(callback: CallbackQuery):
    kb = InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(text="1 🎲", callback_data="coinflip_bet:1"),
            InlineKeyboardButton(text="5 🎲", callback_data="coinflip_bet:5"),
        ],
        [
            InlineKeyboardButton(text="10 🎲", callback_data="coinflip_bet:10"),
            InlineKeyboardButton(text="25 🎲", callback_data="coinflip_bet:25"),
        ],
        [InlineKeyboardButton(text="🏠", callback_data="games_back")],
    ])
    await callback.message.answer(
        "🎲 <b>🎲/🎲</b>\n50/50 🎲 x2!\n🎲:",
        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

@router.callback_query(F.data.startswith("coinflip_bet:"))
async def coinflip_bet(callback: CallbackQuery):
    bet = int(callback.data.split(":")[1])
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        if user.balance < to_decimal(bet):
            await callback.answer("❌ insufficient balance!", show_alert=True)
            return

        coin_msg = await callback.message.answer_dice(emoji="🎲")
        won = coin_msg.dice.value >= 4

```

```

user.balance -= to_decimal(bet)
user.xp += XP_PER_GAME

if won:
    win = to_decimal(bet) * 2
    user.balance += win
    net = win - to_decimal(bet)
    result_text = f"🎲 🎲🎲🎲! 🎲 +{win} 🎲🎲🎲"
else:
    net = -to_decimal(bet)
    result_text = f"🎲 🎲🎲🎲! 🎲 -{bet} 🎲🎲🎲"

await _level_up_check(session, user, callback)
session.add(GameHistory(
    user_id=user.id,
    game_type="coinflip",
    bet=to_decimal(bet),
    result=net,
    details=f"won={won}"
))
await log_balance_change(
    session, user, net, "game_coinflip",
    details=f"bet={bet}, won={won}"
)
await session.commit()

await callback.message.answer(
    f"{result_text}\n🎲 🎲🎲🎲: {user.balance}"
)
await callback.answer()

@router.callback_query(F.data == "game_guess")
async def game_guess(callback: CallbackQuery):
    kb = InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(text="1 🎲🎲🎲", callback_data="guess_bet:1"),
            InlineKeyboardButton(text="5 🎲🎲🎲", callback_data="guess_bet:5"),
            InlineKeyboardButton(text="10 🎲🎲🎲", callback_data="guess_bet:10"),
        ],
        [InlineKeyboardButton(text="🎲 🎲🎲🎲", callback_data="games_back")],
    ])
    await callback.message.answer(
        "🎲 <b>🎲🎲🎲 🎲🎲🎲</b>\n\n🎲🎲🎲 1-6 - x5 🎲🎲🎲!\n\n🎲🎲🎲:",
        parse_mode="HTML",
        reply_markup=kb
    )
    await callback.answer()

@router.callback_query(F.data.startswith("guess_bet:"))
async def guess_bet_start(callback: CallbackQuery, state: FSMContext):
    bet = int(callback.data.split(":")[1])
    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)
        if not user:
            await callback.answer()
            return
        if user.balance < to_decimal(bet):
            await callback.answer("🎲 🎲🎲🎲🎲🎲 🎲🎲!", show_alert=True)
            return

    await state.update_data(guess_bet=bet)
    kb = InlineKeyboardMarkup(inline_keyboard=[
        [
            InlineKeyboardButton(text="1", callback_data="guess_num:1"),
            InlineKeyboardButton(text="2", callback_data="guess_num:2"),
            InlineKeyboardButton(text="3", callback_data="guess_num:3"),
        ],
        [
            InlineKeyboardButton(text="4", callback_data="guess_num:4"),
            InlineKeyboardButton(text="5", callback_data="guess_num:5"),
            InlineKeyboardButton(text="6", callback_data="guess_num:6"),
        ],
    ])
    await callback.message.answer("🎲🎲🎲🎲 🎲🎲:", reply_markup=kb)
    await callback.answer()

@router.callback_query(F.data.startswith("guess_num:"))
async def guess_num(callback: CallbackQuery, state: FSMContext):
    guess = int(callback.data.split(":")[1])
    data = await state.get_data()
    bet = data.get("guess_bet", 1)

    async with async_session() as session:
        user = await get_user(session, callback.from_user.id)

```

```

    if not user:
        await callback.answer()
        return
    if user.balance < to_decimal(bet):
        await callback.answer("❌ 余额不足!", show_alert=True)
        return

    dice_msg = await callback.message.answer_dice(emoji="🎲")
    actual = dice_msg.dice.value

    user.balance -= to_decimal(bet)
    user.xp += XP_PER_GAME

    if guess == actual:
        win = to_decimal(bet) * 5
        user.balance += win
        net = win - to_decimal(bet)
        result_text = f"🎉 猜对了 {actual}! 赢了 {win}!"
    else:
        net = -to_decimal(bet)
        result_text = f"❌ 猜错了 {actual}, 输了 {bet}."

    await _level_up_check(session, user, callback)
    session.add(GameHistory(
        user_id=user.id,
        game_type="guess",
        bet=to_decimal(bet),
        result=net,
        details=f"guess={guess}, actual={actual}"
    ))
    await log_balance_change(
        session, user, net, "game_guess",
        details=f"bet={bet}, guess={guess}, actual={actual}"
    )
    await session.commit()

    await callback.message.answer(
        f"{result_text}\n👤 余额: {user.balance}"
    )
    await state.clear()
    await callback.answer()

@router.callback_query(F.data == "games_back")
async def games_back(callback: CallbackQuery):
    await callback.message.answer(
        "🎮 游戏菜单:",
        reply_markup=games_menu_keyboard()
    )
    await callback.answer()

# =====
# TOPS
# =====
@router.message(F.text == BTN_TOPS)
async def btn_tops(message: Message):
    async with async_session() as session:
        user = await get_user(session, message.from_user.id)
        if not user:
            return
        if not await require_nickname(message, user):
            return
    await message.answer(
        "🏆 <b>排行榜</b>",
        parse_mode="HTML",
        reply_markup=tops_menu_keyboard()
    )

@router.callback_query(F.data == "top_uploaders")
async def top_uploaders(callback: CallbackQuery):
    async with async_session() as session:
        rows = (await session.execute(
            select(User, func.count(Video.id).label("cnt"))
            .join(Video, Video.uploader_user_id == User.id)
            .where(Video.status == "approved")
            .group_by(User.id)
            .order_by(desc("cnt"))
            .limit(10)
        )).all()

    text = "🏆 <b>上传者排行榜</b>\n\n"
    medals = ["🥇", "🥈", "🥉"]
    seen, rank = set(), 0
    for u, cnt in rows:
        if u.id in seen:
            continue

```

```

        seen.add(u.id)
        rank += 1
        icon = medals[rank - 1] if rank <= 3 else f"{rank}."
        text += f"{icon} {get_display_name(u)} - {cnt} ██████\n"
    if not rows:
        text += "██████"
    await callback.message.answer(text, parse_mode="HTML")
    await callback.answer()

@router.callback_query(F.data == "top_viewers")
async def top_viewers(callback: CallbackQuery):
    async with async_session() as session:
        rows = (await session.execute(
            select(User, func.count(VideoView.id).label("cnt"))
                .join(VideoView, VideoView.user_id == User.id)
                .group_by(User.id)
                .order_by(desc("cnt"))
                .limit(10)
        )).all()

    text = "■ <b>████ ██████████</b>\n\n"
    medals = ["■", "■", "■"]
    seen, rank = set(), 0
    for u, cnt in rows:
        if u.id in seen:
            continue
        seen.add(u.id)
        rank += 1
        icon = medals[rank - 1] if rank <= 3 else f"{rank}."
        text += f"{icon} {get_display_name(u)} - {cnt} ██████████\n"
    if not rows:
        text += "██████"
    await callback.message.answer(text, parse_mode="HTML")
    await callback.answer()

@router.callback_query(F.data == "top_levels")
async def top_levels(callback: CallbackQuery):
    async with async_session() as session:
        users = (await session.execute(
            select(User).order_by(desc(User.xp)).limit(10)
        )).scalars().all()

    text = "■ <b>████ ███ XP</b>\n\n"
    medals = ["■", "■", "■"]
    seen, rank = set(), 0
    for u in users:
        if u.id in seen:
            continue
        seen.add(u.id)
        rank += 1
        icon = medals[rank - 1] if rank <= 3 else f"{rank}."
        text += f"{icon} {get_display_name(u)} - ███.{u.level} ({u.xp} XP)\n"
    if not users:
        text += "██████"
    await callback.message.answer(text, parse_mode="HTML")
    await callback.answer()

@router.callback_query(F.data == "top_richest")
async def top_richest(callback: CallbackQuery):
    async with async_session() as session:
        users = (await session.execute(
            select(User).order_by(desc(User.balance)).limit(10)
        )).scalars().all()

    text = "■ <b>████ ██████████</b>\n\n"
    medals = ["■", "■", "■"]
    seen, rank = set(), 0
    for u in users:
        if u.id in seen:
            continue
        seen.add(u.id)
        rank += 1
        icon = medals[rank - 1] if rank <= 3 else f"{rank}."
        text += f"{icon} {get_display_name(u)} - {u.balance:.2f} ██████\n"
    if not users:
        text += "██████"
    await callback.message.answer(text, parse_mode="HTML")
    await callback.answer()

# =====
# QUESTS
# =====
@router.message(F.text == BTN_QUESTS)
async def btn_quests(message: Message):

```

```

async with async_session() as session:
    user = await get_user(session, message.from_user.id)
    if not user:
        return
    if not await require_nickname(message, user):
        return

    today = datetime.utcnow().date()
    quests = (await session.execute(
        select(DailyQuestProgress).where(
            DailyQuestProgress.user_id == user.id,
            DailyQuestProgress.quest_date == today
        )
    )).scalars().all()

    if not quests:
        quest_list = DAILY_QUESTS.copy()
        if is_vip(user):
            quest_list += PREMIUM_DAILY_QUESTS
        for q in quest_list:
            session.add(DailyQuestProgress(
                user_id=user.id,
                quest_type=q["type"],
                quest_date=today,
                progress=0,
                target=q["target"],
                reward=to_decimal(q["reward"]),
            ))
        await session.commit()
        quests = (await session.execute(
            select(DailyQuestProgress).where(
                DailyQuestProgress.user_id == user.id,
                DailyQuestProgress.quest_date == today
            )
        )).scalars().all()

    text = "■ <b>■■■■■■■■■■ ■■■■■■</b>\n\n"
    for q in quests:
        status = "■" if q.completed else "■"
        claimed = " ✓ ■■■■■■■■" if q.reward_claimed else ""
        text += (
            f"{status} {q.quest_type}: "
            f"{q.progress}/{q.target} - "
            f"{q.reward} ■■■■■■{claimed}\n"
        )

    await message.answer(
        text,
        parse_mode="HTML",
        reply_markup=quests_keyboard(quests)
    )

@router.callback_query(F.data.startswith("quest_claim:"))
async def quest_claim(callback: CallbackQuery):
    quest_id = int(callback.data.split(":")[1])
    async with async_session() as session:
        quest = (await session.execute(
            select(DailyQuestProgress).where(DailyQuestProgress.id == quest_id)
        )).scalar_one_or_none()
    if not quest:
        await callback.answer("■■■■■ ■■ ■■■■■■.", show_alert=True)
        return
    if not quest.completed:
        await callback.answer("■■■■■ ■■■ ■■ ■■■■■■■■!", show_alert=True)
        return
    if quest.reward_claimed:
        await callback.answer("■■■■■■■ ■■■ ■■■■■■■■!", show_alert=True)
        return

    user = await get_user_by_id(session, quest.user_id)
    if not user:
        await callback.answer()
        return

    quest.reward_claimed = True
    await log_balance_change(
        session, user, quest.reward,
        "quest_reward", source_id=quest.id
    )
    user.balance += quest.reward
    await session.commit()

    await callback.answer(f"■ ■■■■■■■■ {quest.reward} ■■■■■■!", show_alert=True)

@router.callback_query(F.data.startswith("quest_done:"))
async def quest_done(callback: CallbackQuery):

```

```

await callback.answer("■ ■■■■■ ■■■■■■■■, ■■■■■■■ ■■■ ■■■■■■■■.", show_alert=True)

@router.callback_query(F.data.startswith("quest_info:"))
async def quest_info(callback: CallbackQuery):
    await callback.answer("■ ■■■■■ ■■■ ■■ ■■■■■■■■. ■■■■■■■■■■■■!", show_alert=True)

# =====
# HELPER: ■■■■■■■■■■ ■■■■■■■■
# =====
async def _update_quest_progress(
    session,
    user_id: int,
    quest_type: str,
    amount: int = 1
):
    today = datetime.utcnow().date()
    quests = (await session.execute(
        select(DailyQuestProgress).where(
            DailyQuestProgress.user_id == user_id,
            DailyQuestProgress.quest_type == quest_type,
            DailyQuestProgress.quest_date == today,
            DailyQuestProgress.completed == False
        )
    )).scalars().all()

    for quest in quests:
        quest.progress = min(quest.progress + amount, quest.target)
        if quest.progress >= quest.target:
            quest.completed = True

    try:
        await session.commit()
    except Exception:
        await session.rollback()

=====
FILE: requirements.txt
=====
aiogram==3.15.0
aiohttp>=3.9.0,<3.11
sqlalchemy==2.0.36
aiosqlite==0.20.0
asyncpg==0.29.0
python-dotenv==1.0.1
reportlab==4.2.2

=====
FILE: TODO.md
=====
# Bug Fixes - Print to Logger Cleanup

## Progress
- [x] Analyze codebase and identify bugs (prints, no logger)
- [x] Create edit plan and get approval

## Tasks
- [x] Edit app/make_project_pdf.py: Replace all print() with logger calls, add setup_logging(), wrap main() try/except
- [x] Edit app/user_handlers.py: Replace print("\nSTART COMMAND RECEIVED\n")
- [x] Edit app/migrate.py: Replace print("\nMigrations applied successfully\n")
- [x] Test: run python app/make_project_pdf.py (check PDF + logs no prints) - SUCCESS: PDF generated with JSON logs, no prints
- [x] Test: python app/migrate.py (logs) - SUCCESS: Logger setup, runs (migrate error expected if no DB tables)
- [ ] Run bot, /start (no print)
- [ ] attempt_completion

**Estimated fixes: 3 files, cleanup for prod-ready**

```